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注: 此卡片主要作为外购类产品零部件图样的临时代替品, 在暂无相关设计图样, 但确需进行常规采购时编制。

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Product Manual

XO5339Z336

Adjusting Manual(CON4342D)

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HANGZHOU SWORD ELEVATOR CO.,LTD.

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Debugging Instructions

No.: _____

Name: _____

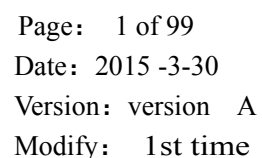
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Date: 2015.3.30

Hangzhou Sword Elevator Co., Ltd

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HAMCB board rapid debugging method:

1. All the mechanical part installation complete, no short wires in control cubicle;
2. Make sure every safety switch of safety circuit works, and lock of landing door on each floor effective.

Observe the indicator on HAMCB electronic board, and check whether the input signal is correct:

Indicator	Normal State	Descriptions
RUN	Flash	Flash: remote serial line and watchdog normal
+5V	Light up	Light up: power supply (5V) normal
ES	Light out	Light up : emergency stop button action
DW	Light up	Light up : landing door close
DCL	Light up	Light up : reach the door open limit switch (front door and rear door)
DOL	Light out	Light up : reach the door open limit switch (front door or rear door)
DZ	Light up when leveling	Light up : lift car in door zone
DFC	Light up	Light up : car door and safety chain closed
DOB	Light out	Light up : door reversal device(front door or rear door) is operated

Note: if the indicator state is inconsistent with the state as listed in the table after power on, please check the relevant circuit and parameters (normally parameters are set up before out of elevator factory).

3. Use ERO or TCI to run the elevator and check 1LV and 2LV in door area. make a forced slowdown of hoist way signals including 1LS and 2LS: Check DZ light of mainboard or test voltage of input port by multimeter.
4. Follow the simple debugging step of hoist way self-learning on the control cubicle. Or complete the following steps.
5. Verify RUN ENABLE parameter of M-3-3-1-9 in ALMCB driver is 0, and RUN SOURCE SVT of M3-3-5 is 1.
6. Put ERO and TCI to the normal position, and use the server to operate the driving part of M-3-3-4 to start the hoist way self-learning.
7. Elevator runs at a low speed to the bottom leveling position, and run up at self-learning speed for hoist way self-learning, after reaching the top leveling floor, self learning is completed.
8. After self learning is completed, change the RUN ENABLE parameter to 1. The information of hoist way location is stored in the M-3-1-4 menu, please check whether it is correct.

Counterbalancing coefficient check

Verify if counterbalancing coefficient meets 40%~50% as required usually is 45%.

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Put the balancing load (45% of the rated load) into the car, monitor Output current in M3-1-1, record the value of elevator on the intermediate floor in the process of ascending and descending, and the recorded current value ascending and descending should be equal (less than 1A in difference), or increase and reduce counterweight filler on the elevator counterweight frame, increase the counterweight if ascending current is big, otherwise, reduce the counterweight.

Or manually operate the jigger, see if it's balance by the left and right strength of force.

The lift weight load control instrument adjustment

Before the adjustment, empty capsules need to keep at least half an hour. Requires constant adjustment process, cannot interrupt.

A control instrument installation:

Controller installed in the elevator car at the bottom

A) contour diagram:

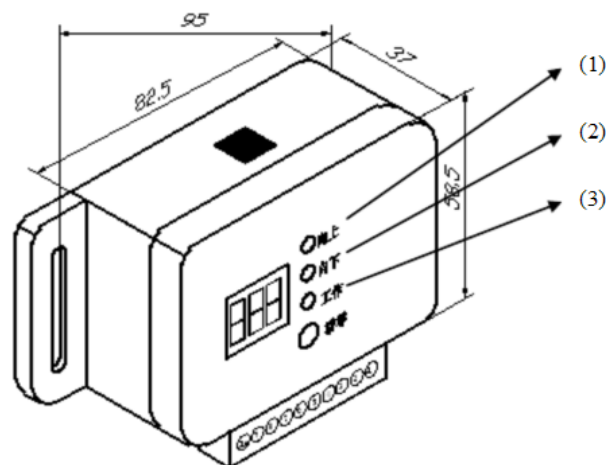


Figure 1

1. The position reference indicator light (1) : 17 mm when $\leq$ clearance, the lights (should be adjusted upward);
 2. The reference position indicator (2) : when $0 < \leq$ clearance = 14 mm, the lights (should be adjusted downwards);
 3. The working indicator light (3) : normal work when the lights; Since learning, used to indicate correct and study
- B) installation schematic diagram:

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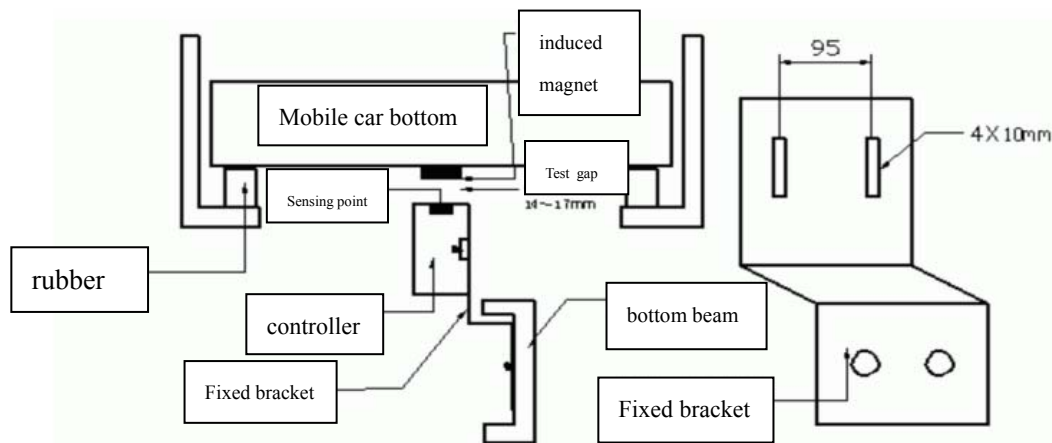


Figure 2: installation diagram

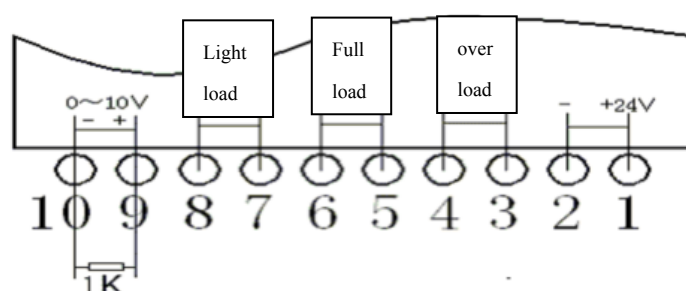
1. Under the condition of car racing, the magnet adsorption at the center of the bottom, and under the logo facing;
2. Use M3 bolts will control device is installed on the fixed bracket, then fixed bracket in capsules position is corresponding to the magnet on the sill, make prompt induction center on controller for induction magnet center position, ensure the magnet on the logo with the logo on the control instrument for strictly, and the controller is parallel with the magnet.
3. Adjust the control instrument testing gaps for about 15 mm, if the installation position is not correct, when installing two position indicator is on, the indicator (1) on the upward control instrument need to be adjusted upward, downward indicator 2 on the downward adjustment control instrument, location, just two position reference light is completely destroyed.
4. The actual installation, if the bottom displacement range for more than 15 mm, could detect clearance will be increased to more than 15 mm, but now light near the output signal will be offset. This installation is not recommended.

C) accessories:

1. Control instrument fixed bracket need to users according to the specific circumstances of homemade;
2. The controller USES a magnet for special 30 * 30 * 53 mm square strong magnet;
3. Suggest choosing from no load to load the bottom displacement at about 6 ~ 8 mm rubber piece.

Second, the connection details:

Four control instrument with a total of 10 terminals, as shown in figure 3,



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Figure 3: connecting terminal

The terminal USES respectively

- I. 1, 2, DC24V power supply for the control instrument, 1, 2 is negative;
- Ii. 3, 4 for the overload relay contact, contact for work still open points;
- Iii. 5, 6, is full of relay contacts, contact for work move close point;
- Iv. 7, 8 for light load relay contact, contact for work move close point;
- V. 9 or 10 for instrument output within 0 ~ 10 ma current signal or 0 ~ 10 voltage signal, 9 is positive, 10 is negative.

1. Install the controller in capsules palankeen, light to electricity, the three digital tube display "821";
 2. The elevator no-load reset. Press the reset button three times in a row, work lights out, accompanied by three light rings at the same time, about 5 seconds to hear three long response, and the work lights to light up, zero after learning, digital tube display "000";
 3. Increasing load to the need of light load and within 5 seconds press the reset button five times in a row, with 5 short rang, work lights out at the same time; About 5 seconds long heard a loud ringing, and work lights to light up, show that light load point after learning;
- Deadweight are then added to the amount of load value, press the reset button, along with a short rang and work light is put out at the same time, long press 5 seconds later, heard a long echoes, and in a short work lamp lit again, show full finished learning, digital tube display "100".

Upgoing overspeed protection test operation

1 testers in room test uplink speeding, normal operation of the elevator to the bottom, and placed in the DDO, CHCS state.

2 normal operation of the elevator to the bottom, put ERO in maintenance state, open the speed limiter cover, easy to switch to directly observe the speed limit

Movement and recovery.

3 modification parameter, make the elevator running faster than speed limiter action speed.

At rated speed 0.4 m/s, the host rated speed 153 RPM, for example, assume that at the rate of 115% of the rated speed, speed limiter action

Change the following parameters to make the normal operation of the elevator speed 0.48 m/s (120%).

In the HAMCB board and with the server set the following parameters:

< 3-1-4 M3 - > Duty Speed * 1.2 = the rated Speed

< 3-1-4 M3 - > Rated RPM = Rated speed * 1.2

Rated FRQ (Hz) * 1.2 = Rated the original host frequency

< 3-1-9 M3 - > Velocity normal = Velocity normal * 1.2

4 maintenance running up and down the elevator, after the maintenance operation is normal, the elevator maintenance operation to flat bottom layer position, and will repair ERO

To the normal running state.

5 with a server operating the elevator, elevator uplink speeding test (1).

Note: the input operation instruction floor can reach speeds speed limiter action can, suggested input floor is as follows:

Rated the elevator was 0.4 m/s elevator, input instruction floor should be less than or equal to (N - 1).N to the top floor.

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Rated the elevator was 1.0 m/s elevator, input instruction floor should be less than or equal to (N - 1).N to the top floor.

Rated the elevator elevator for 1.5 ~ 1.75 m/s, input instruction floor should be less than or equal to (N - 2).N to the top floor.

Rated the elevator was 2.0 m/s elevator, input instruction floor should be less than or equal to (N - 5).N to the top floor.

6 test finished, put the elevator in ERO maintenance state, restore the parameter Settings, and make the necessary inspection, to ensure normal operation of the elevator.

7 if you have the need of multiple tests, please repeat the above steps.

Safety gear linkage test

Normal operation of the elevator to the bottom, and placed in the DDO, state of CHCS, 100% rated load weight is put inside capsules, will control the ERO contained in the maintenance switch to maintenance position, elevator maintenance speed up operation first layer;People make the speed limiter action, elevator running down, while speed limiter electrical switch should be;Continue to run down the elevator, safety gear, its top safety clamp switch action;Run down, continue to repair speed governor will lift the car safety wire rope clamp, put the brakes on the car, at this time should be observed in wire rope traction wheel skidding or wire rope relaxation.

Light on the traction test

Elevator no-load running to the top, will control the ERO contained in the maintenance switch to maintenance position, with access speed upward run, when the pressure in the bumpers and traction machine elevator upward direction, should be rope slipping or cannot ascend no-load capsules.

DBD test

Cut off the current brake, consists of two independent BY and SW main contactor.When the elevator stops, if one of the contactor's main contact is not open, DBD faults on frequency converter, the elevator will not run this again.

DDP test

Traction drive elevator limiter is equipped with the motor running time, in the following cases to keep the elevator drive host stop running and in stop state:

When start the elevator traction machine does not turn;

Capsules or heavy downward movement due to obstruction, traction rope in traction wheel skidding.Limiter motor running time should be no greater than the following two time values smaller value of the work:

1) 45 s;

(2) the elevator running full time plus 10 s.If run time less than 10 s of the distance, the minimum value of 20 s.

Calibration program is designed to run normally, running, running on a flat layer near the biggest photoelectric the elapsed time between two flat layer

Upper and lower limit test

Limit switch installed distance end station reliability in 100-200 mm, flat layer connection correct, ERO can receive short run, limit switch should be in the car or the weight (if any) before work contact with buffer, and continues in its state of motion during the buffer is compressed.

Upper and lower limit test

Elevator maintenance 2 ls and run to the end station 1 ls/dz activity is limited, but can run in the opposite direction,

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the limit switch can't action, parameters of ERO - TYP set to 1

The brake switch test

Brake air gap adjustment to 0.25-0.4 mm, switch wiring are in conformity with the drawings, detection open, reliable. Will open the elevator to the top floor of the elevator to switch to the maintenance state, by adjusting the switch plunger screw until: use 0.1 MM feeler into the upper brake air gap, manual wrench loose brake, the brake switch action, with a 0.15 MM feeler into the upper brake air gap, manual wrench loose brake, the brake switch not action.

Forced to slow down switch test

Switch base from the bow to face distance 10 + 2 mm, switch track in the middle of the bump bow position, switch direction and run in the same direction, installation position and the instruction manual, correction and function of normal operation and reliable running, by modifying the Decelera normal than Decelera NTSD 100, the whole operation not stampings, trouble-free. After the completion of the test value to change back to the factory.

ERO run

ERO emergency electric lines, upper and lower limit switch can be short, buffer switch, limit switch, safety clamp switch, cannot meet the urgent stop switch short, when tight round switch, convenient to implement the rescue, safety gear linkage test, traction on the no-load test.

Braking force self-monitoring

- Braking force manual monitoring function; Manual M-3-1-6, confirm inspection and empty load, start braking force monitoring.
- Braking force self-monitoring function; Parameter M1-8-10, BrkT-D(Self detect date and week, default 15 parameter range 10-15), BrkT-T(Self detect absolute hour, default 00, parameter range 0-24), BrkT-M (Self detect absolute minute, default 00, parameter range 0-60)

Remarks: Brake switch monitoring function shouldn't be blocked.

Car unintentional movement protection

Manual M137, pass code 7588, input up or down command (Car upward unintentional movement test input up command, car downward unintentional movement input down command), when door close, input present floor command and make the door open; when door open, press blue button+2, then it will start UCM test. Reset UCM fault method as bellow: Blue button + 0 or INS+TOP+BOT button should work simultaneously.

Remarks: When test it must make sure there is advanced door opening function, and with LVCT1 safety board. Elevator door opening floor should have special door keeper to watch, "KEEP OFF".

Steel belt bending time and production date monitoring instruction

- Steel belt production date monitoring, when M1-3-1-6 last group time setting to be 0, (Suitable to elevator type except XO-GMEIII、XO-GMEIII-MR、WD800MRL、WD800MR), setting production year and month(example: 201604, suitable for XO-GMEIII、XO-GMEIII-MR、WD800MRL、WD800MR), start monitoring function, when time expire, trigger SE fault, and cannot restore 0 setting. Production date can only change 5 times.

Remarks: Steel belt production date is marked on steel belt, pls pay attention.

- Steel belt bending time monitoring, monitoring manual M3-1-7, only when traction sheave diameter is

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less than 150mm, trigger bending time monitoring, when over than 10 million times, it will have SE fault.

Door lock short circuit protection function

DLM (Door lock short circuit protection)mode is effective, cannot be changed; detailed parameter M1-8-10 NoDW_Chk=0.

Function description:

1.DW is short circuit; 2.DFC is short circuit; 3. DW and DFC is short circuit, enter DLM mode, try to open or close door 3 times, door open waiting time according to normal door open waiting time. In the whole process all hall call command is reserved but not reacted.

If detect signal is not short circuit in all 3 times, then exit DLM mode.

If 3rd time detect is short circuit, after finish then indicate“Door Bridge” blinking. Then all hall command cancelled and cannot register other command.

The 4th time door open and let passenger out, setting door open waiting time according to DAR-T. After door full close, when cabin DOB and DOS is effective, can open the door, after wait DAR-T time and close the door. LRD, EDP can open door effectively during door close.

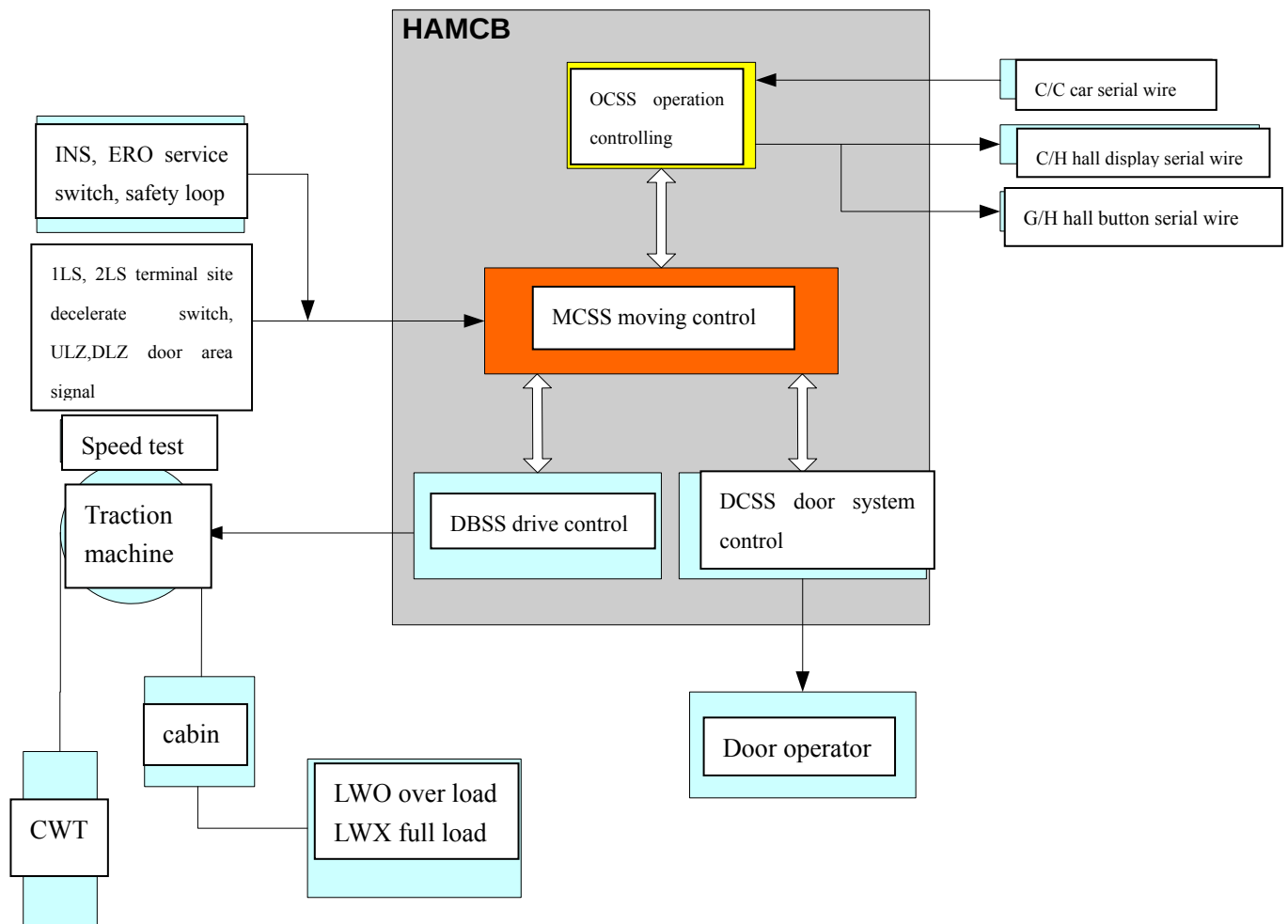
At this time, by ERO/TCI OFF->ON, non-door zone elevator should inspect move to door zone, ERO/TCI ->OFF, then elevator door open and check car door and landing door lock state, if normal, then elevator back to normal mode. If not (short circuit), wait DAR-T time and door close.

Power off and RESTART, keep DLM mode.

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overview

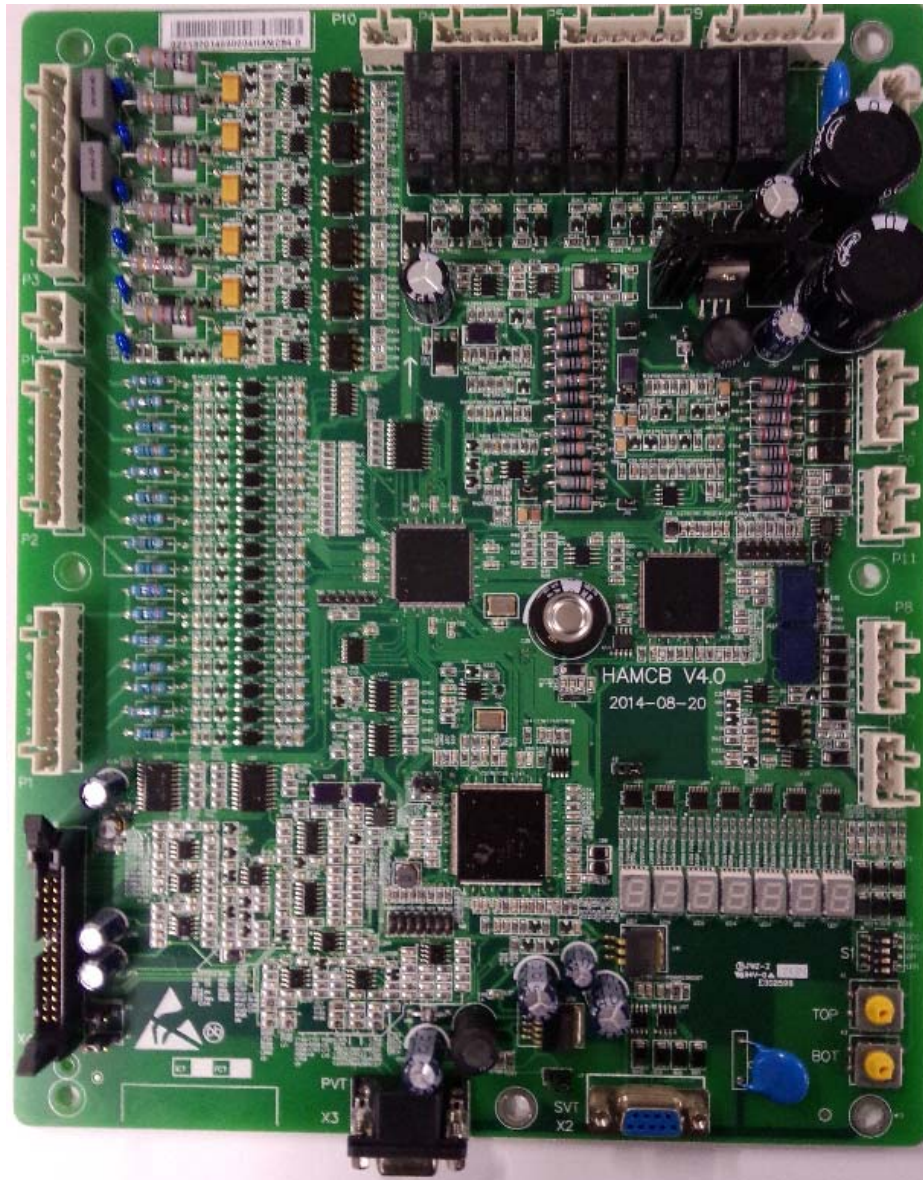
HAMCB board is the core part of the control system, according to the modular design, contains the OCSS (control) operation, MCSS (operation control), DBSS (driver) and DCSS (door control system).



Elevator system structure diagram

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1.1 Product pictures

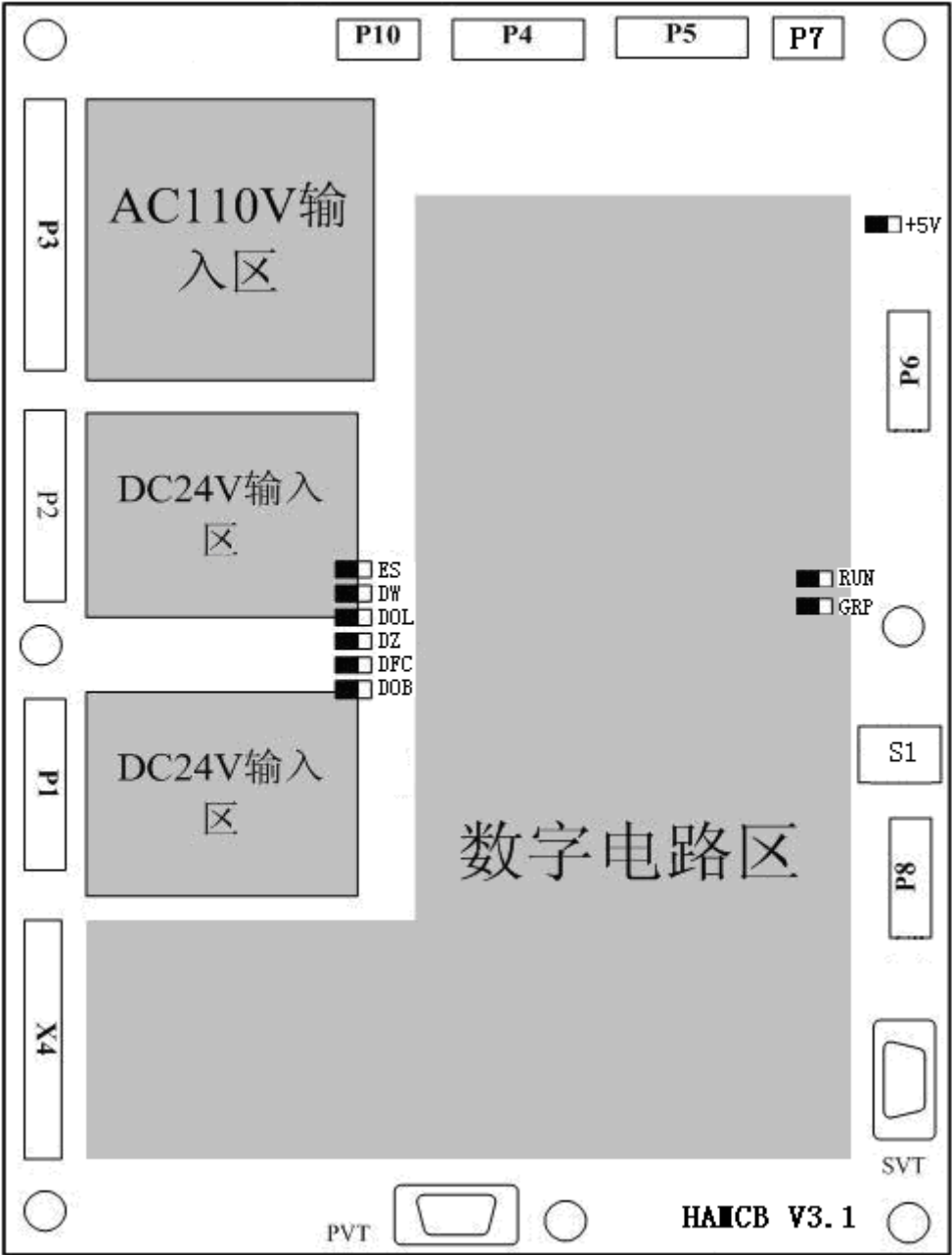


1.2 product structure

1.2.1 HAMCB distribution circuit diagram

HAMCB board circuit is divided into: analog circuits and digital circuits, analog circuits include: power supply circuit, communication circuit, input and output circuit, the server interface circuit, encoder circuit, analog detection circuit; Digital circuit includes: CPU circuit, CPLD circuit and other peripheral

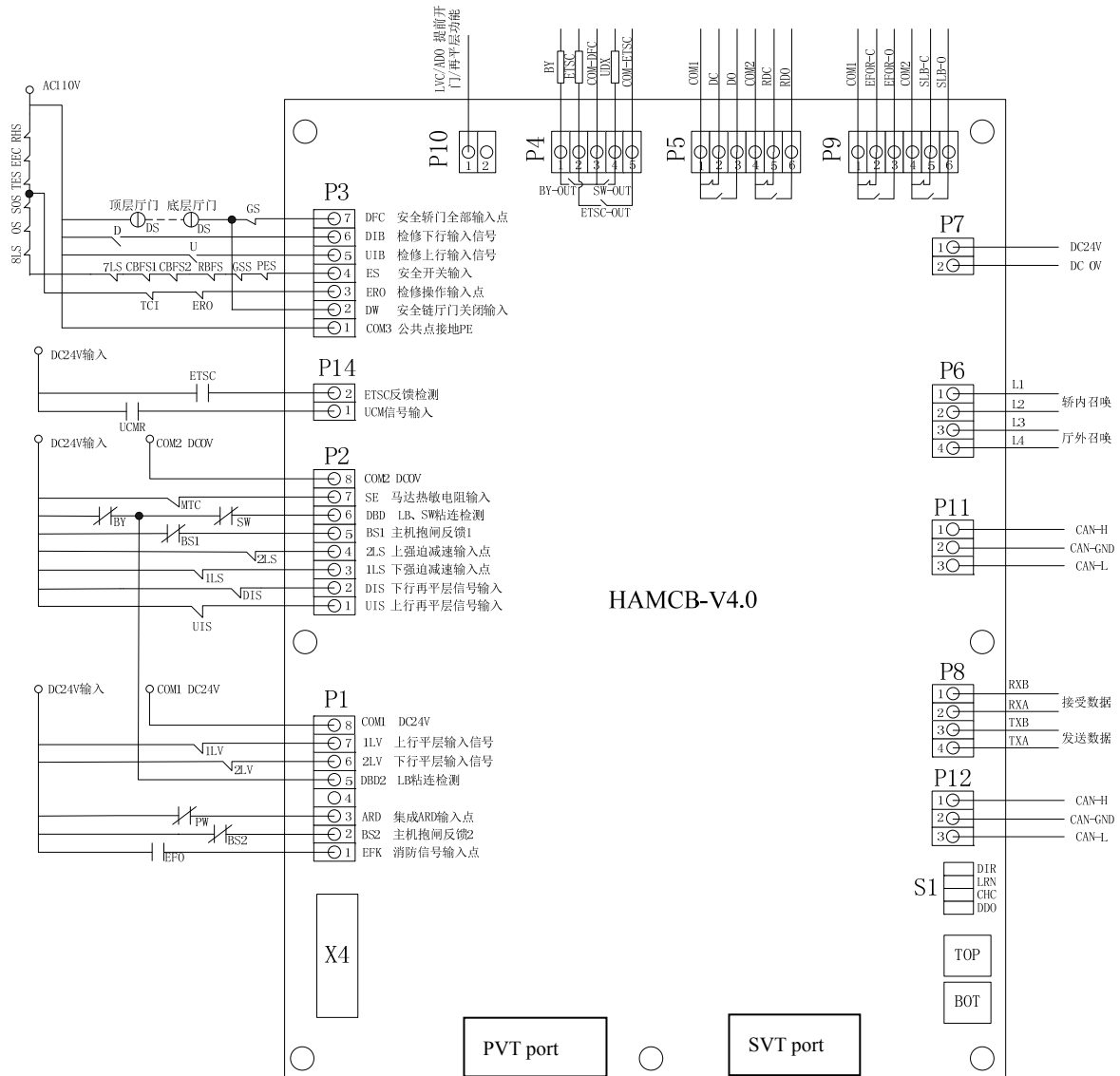
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circuit;

HAMCB board

1.2.2 HAMCB outer wiring diagram



Outer wiring diagram

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1.2.3 HAMCB plate buttons, dial the code switch, light and port

1.2.3.1 HAMCB plate dial the code switch

code	function	notes
TOP	top floor command	
BOT	bottom floor command	

code	function	notes
DDO	Open the closing operation that can/ban	
CHCS	Outside the hall call operation can make/ban	
DIR	The elevator running direction change switch	
LRN	Hoistway self-learning start switch	

1.2.3.2 HAMCB board indicator light description

code	function	notes
+5VL	5 v power supply is normal	
RUN	Blink means communication is normal	
GRP	On: means in the group control state Off: means not in a state of group control	
RSL	Blink means normal communication	
VLC	5 v power supply is normal	
DZ	On: means mun signal effectively Off: means mun signal is invalid	
DOB	On: means have to open the door Off: means no open signal	
DOL	On: means the door in place signal effectively Off: means the door in place signal is invalid	
DCL	On: means closed in place signal effectively Closed off: means in position	

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	signal is invalid	
DFC	On: means the car door shut Off: means the car door open	
DW	On: means hall door closed Off: means hall door open	
ES	On: means safety circuit disconnection Off: means the safety circuit conduction	
INS	On: means in a state of repair Off: means not in a state of repair	
NOR	On: means in the normal state Off: means not in the normal state	

2. Debug instructions

2.1 maintenance mode condition check

Tip: before moving slow train, make sure all the mechanical parts have been debugging is completed, please refer to installation instructions.

2.1.1 check control cabinet

Open the door of the control cabinet, check for joint looseness and component damage, keep random data, the replacement of damaged parts, tighten control all joints in the cupboard. Please pay special attention to the power cord, power line and fastening braking resistance wire connection.

2.1.2 wiring check

According to the wiring diagram, check PVT line, accompanying the interim of the cable connection, limit switch, temporary, check whether each device grounding line and reliable grounding.

2.1.3 insulation inspection

OFF the ground and the connection of HL, pull out all the plugin on the HAMCB, all mun, summon relevant plug-ins, instructions, and signaling pull out, put all the air switch in "OFF" position, with insulation meter measuring ground and HL, power cord, motor power line, safety circuit, control circuit, the brake coil, door machine, lighting on both ends of the insulation resistance value, to ensure that the insulation resistance value within a specified value, back in the ground and HAMCB plug-in.

circuit	Allowable resistance	insulation
Power circuit and safety circuit	$\geq 0.5M\Omega$	
Control circuit (including door machine), lighting circuit and signal circuit	$\geq 0.25M\Omega$	

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Note: in testing insulation resistance, be sure to get off the plug on the electronic board, otherwise may damage the electronic board.

2.1.4 check the input voltage

To cut off the main power switch and control air incase of other air switch, check whether the single phase input voltage is within the specified range (+ / - 10%), and according to the actual input voltage regulation transformer input terminal of the connection, check the open CPB CPB - 2, CPB - 4 if voltage (lighting) voltage is 220 v + 10%, check the open the CP1 voltage (power supply transformer and gantry crane) is 220 v + 10%, check whether the power light on the driving components normally (light is below HAMCB plate hole).

2.1.5 check control transformer output voltage

Close the main power supply air switch, check whether the voltage of the transformer output and drawings consistent (in the case of the input voltage meet the requirements. Transformer output have $\pm 10\%$ error) 。

2.2 electrical inspection

2.2.1 check HAMCB input and output voltage

Cut off the power and pull HAMCB plate all the plug-ins, and then close the air switch CP4, HAMCB plate is measured with a multimeter P7 plugin of 1, 2 feet, check if the voltage is DC24V plus or minus 10%, if not, please check the 220 v output transformer.

2.2.2 on electric check HAMCB state

Cut off the main power switch, plug in all plug-ins.

Check HAMCB electronic board on the DDO/CHCS/DIR/LRN) switch position.

Dial the code switch has the following features:

(DDO) - ON: normal operation;

- OFF: cancel the door operation.

(CHCS) - ON: normal operation;

- OFF: cancel the hall outside the call.

(DIR) - ON: host of right;

- OFF: the host left;

(LRN) - ON: normal operation;

- OFF: hoistway self-learning switch.

Short of caking J1, J2, J3, J7, J8 and J10, J11, J12, J15 and J16 has the following features:

J1 U11 chip program burning port

Write protect switch J2 drive parameters

J3 U1 chip program burning port

J7 encoder power selection (5 v, 8 v)

J8 J10 J15 J16 communication terminal to absorb to choose

J11 J12 encoder type choice, short answer on the right side of the two pin type using incremental

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encoder;Short after the left two pins

Using cosine encoder

Server interface SVT has the following features:

SVT server interface

Confirm control ark ERO switch the emergency operation of the position.

Confirm all the hall door and the car door is fully closed.

Close the main power switch.

Observe HAMCB the indicator lights on the electronic board, check whether the input signal is correct:

Indicator light	Description
RUN	Blink: normal remote serial line and guard dog
+5V	On: the power supply is normal (5 v)
ES	On: stop button action
DW	On: the hall door closed
DCL	On: to close the door in position switch (the front door and back door all in place)
DOL	On: to open the door in position switch (the front or the back door)
DZ	On: capsules in the door zone
DFC	On: 9501 and safety chain closed the car door
DOB	On: reverse device (the front or the back door) by operation

Note: if after power on, the state of the light and the state is not listed in the table, please check the relevant circuit and parameters (usually parameter is set at the launch of the elevator factory).

3. HAMCB board parameter Settings

Within 3. HAMCB all installation and I/O parameters when the factory has been set up, details please refer to HAMCB related parameter list.If necessary, to confirm the following parameters:

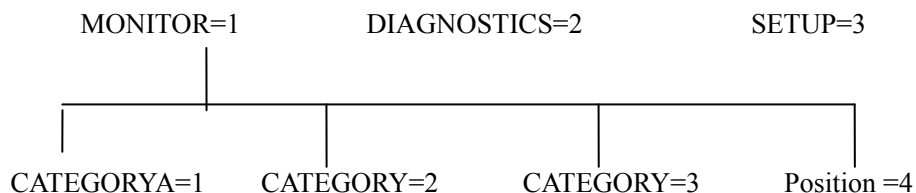
parameters	Description	Set value	notes
M-1-3-1-1			
TOP	Top floor	In accordance with contract	Start from 0
BOTTOM	Bottom floor	0	
M-1-3-1-2			
DRIVE	Driver type	0	If any releveled and open early function is set to 1
LV-MOD	Leveling mode	0	
DZ-TYP	Door region type	1	
M-1-3-1-3			

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DOOR	Door operator type	5	
F: EN-DCL	DCLSignal input to define	1	
F: DOL-D	DOLSignal input to define	0	
M-1-3-1-4			
NoDW_Chk	DW signal test	0	

4. The drive part parameter Settings

4.1 Insert the SVT interface, press 4 to enter the password Settings menu;The driver module level 1 menu press 3, specific display as follows:



Monitoring drive state press 1, enter the 2nd level menu above motioned in monitoring:

parameters	parameters description	monitoring value	notes
M-3-1-1			
Software version	Software version		The encoder feedback speed
Car speed mm/s	The elevator actual speed		
Motor Speed RPM	Based on motor speed (PG)		
Rotor position	Rotor position Angle (mechanical)	0~359.9	
Dictated vel mm/s	Speed for a given (DBSS)		The given speed
Output voltage	The output voltage of a given	0~900V	
Mtr trq PU	Torque current is given (relative amount of torque current)	1.0—>The amount of torque	1.000
Output current	The output current	0~999.9A	The actual detected output current
Heat sink temp1	The radiator temperature of 1	actual temperature	
Heat sink temp2	The radiator 2 temperature	actual temperature	

parameters	parameters description	monitoring value	notes
M-3-1-2			
DC link V	Dc bus voltage	Actual voltage	

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X offset (A)	X offset phase current sensors	/	The actual current sensor bias
Y offset (A)	Y offset phase current sensors	/	
Z offset (A)	Z offset phase current sensors	/	
Frequency (Hz)	The output frequency		
Speed gap (mm/s)	Speed gap		
LS distance (mm)	Forced slowdown should install distance (corresponding to set the parameters)		
Motor overload	Motor overload integrator		
Drive overload	The inverter overload integrator		

parameters	parameters description	monitoring value	notes
M-3-1-3			
Encoder pulse	The encoder pulse condition	0~65535	
Encoder Z num	The encoder Z count	0~65535	
Encoder C POS	Encoder C position	0~65535	
Encoder D POS	Encoder D position	0~65535	
Factory resd5{tc \l 4 "5.3.4.9 UDX dropped"}	Manufacturers test 5	0~65535	
NOW FLOOR	The current floor	00	
POSTION	The current position	000.00m	
INPUT1	INPUT1		
INPUT2	INPUT2		
OUTPUT	OUT		
OUT CODE	OUT CODE		
Prepare to run	Ready for operation (manual or to DBSS MCSS)		
Load Precent	The actual weighing compensation amount		
Bases Enabled	DBSS state		

parameters	parameters description	monitoring value	notes
M-3-1-4			
2LS (m)	Forced to slow down on 1 position		The benchmark is 1 m
4LS (m)	Forced to slow down on 2		

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	position		
1LS (m)	Under the force reduction 1 position		
3LS (m)	Under the force reduction 2 position		
	The floor height		

Monitoring the fault record press 2, enter the fault monitoring level 2 menu pictures as follows:

The CURRENT FAULT = 1 SAVE FAULT = 2 CLEAR CURRENT FAULT = 3

CLEAR ALL SAVE FAULT = 4

On the parameters set according to 3, into the parameters set level 2 menu indicated as follows:

Most cerebral sci-film Contract Data = 1 = 2 Pr. Ini = 3 Height learn = 4 LOAD = 5 T.N TSD = 6

Press 1 to enter the parameters setting of 3 menu that shows the following:

The FIELD ADJUST = 1 VEL REGULATOR = 2 BRAKE/MOTOR DBR = 3 = 4

TIMING = 6 ENHANCED = 9 LEVEL = 0

The parameters according to 2 will set menu.

Press 3 will to initialize all of the parameters, please use the menu;

According to 4 will position hoistway self-learning;

For the use of synchronous motor of elevator installation, must carry out automatic positioning of the host and hoistway self-learning. Must be set before the positioning and hoistway study the parameters of motor parameters and encoder, according to the M3-3-1-4 into the motor parameters setting, in accordance with the site specific data using input motor nameplates. Encoder parameters in FIELD ADJUST = 1.

At the scene of the parameters adjusted into (factory value parameters according to 0.4 m/s, 1.2 KW host, for example)M3-3-1-1

key value	display content	description	factory default	value scope
M3-3-1-1	Inertia kg/m2	The system of inertia	2	Note: asynchronous machine set the range of 10 below
	Rotate dir	Direction of rotation	0	0、1
	Pretrq trim (PU)	Pre torque gain	1.00PU	Don't have to
	ENCODER PPR	The encoder pulse number	2048	0~10000
	Encoder sort	The encoder types	1	0: square-wave incremental

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				1: is cosine
	Encoder dir	Direction of the encoder	NA	NA
	Duty speed[mm/s]	Rated speed	Automatic calculation	Are not allowed to set up
	Delay brk lftd[ms]	The brake hold action delay	500ms	0~9999ms
	Delay lft brk[ms]	Open the brake time delay	500ms	0~9999ms
	MTC Sw Pres? 1/0	The motor thermal switch test	1	0、1
	Type of motor	main engine type	0	0、1

The speed regulator parameters adjustment into the M3-3-1-2

key value	display content	description	factory default	value scope
M3-3-1-2	SpdP1 gain	The proportional controller gain 1	1500	invalid
	SpdI1 gain	integral controller gain 1	500	
	ARD Speed (mm/s)	ARD running speed	100	1~500
	ARD Run Direct	ARD running direction	0	0~1
	SGP tran21 thr%	SGP convert 21 threshold value	30	0~100
	SGP tran21 band%	SGP convert 21 bandwidth	10	0~100
	Reset speed %	reset speed	50	1~100
	Feedback filter 1	velocity feedback smoothing 1	300	0~300 Adjust mechanical resonance
	Feedback filter 2	velocity feedback smoothing 2	300	
	Track error mm/s	Speed maximum allowable deviation	500	1~1000
	ETSC present(%)	ETSC protection factor	97	0~100
	ETSC Enable(0/1)	ETSC protection enable	0	0~1
	Overspeed (PU)	overspeed rate	1.10	0~1.5

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M3-3-1-3 is brake switch set

key value	display content	description	factory default	value scope
M3-3-1-3	Brk SW pres	Brake switch test	1	0~1, 0: not testing, 1: testing

electrical machine parameters setting is in M3-3-1-4, according to different main engine nameplate input (e.g.:1.2KWmain engine) :

key value	display content	description	factory default	value scope
M3-3-1-4	Rated power[KW]	rated power	1.2	0.1~999.9KW
	Number of poles	number of machine poles	10	2~100
	Rated RPM	rated revolving speed	153	2~9999
	Rated frq	rated	12.8	0.01~99.99
	Rated voltage	rated voltage	220	10~999
	Duty load[kg]	rated load	400	10~9999
	Rated I[A]	rated current	4.6	0.1~999.9
	Rated Trq (Nm)	rated torque	75	note: when you use asynchronous machine, like main engine, if nameplate doesn't show parameters, set it around 80.

M3-3-1-6=TIMING set timing parameters

key value	display content	description	factory default	value scope
M3-3-1-6	Brk dect dlay ms	Brake testing time	1000	0~2000
	0mm/s t lim (ms)	Zero speed time	0	0~5000

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	Brake settle(ms)	Brake releasing time	1300	0~5000
	APR time(ms)	Starting torque reimbursed time	200	100~900

M3-3-1-9=ENHANCED, set strengthen parameters

key value	display content	description	factory default	value scope
M3-3-1-9	Floor number	Floor number	Contract value	02~55
	Velocity normal	elevator rated speed	400	0~10000
	Accelera normal	acceleration	350	10~1500
	Jerk0 normal	Add acceleration 0	150	0~1500
	Jerk1 normal	Add acceleration 1	250	0~1500
	Decelera normal	deceleration	400	10~1500
	Jerk2 normal	Add acceleration 2	150	0~1500
	Jerk3 normal	Add acceleration 3	200	0~1500
	Velocity inspect	service speed	250	0~640
	Accelera inspect	service acceleration	200	10~1500
	Velocity learn	self learning speed	100	0~500
	Position gain	position gain	20	10~40
	Velocity relevel	Releveling speed	30	0~100
	Accelera relevel	Releveling acceleration	300	10~1500
	Decelera NTSD	NTSD deceleration	1000	10~1500
	Jerk NTSD	NTSD Add acceleration	200	0~1500
	Pos delay	Door area delaying	100	0~300
	Shv diam (mm)	Traction wheel	100	10~10000

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		diameter		
	Gear ratio	reduction ratio	1	001.0~100.0
	Rope ratio	Roping rate	2	01~06
	Run enable	Running enable	0	0, 1

M3-3-1-0=LEVEL, set leveling parameters

key value	display content	description	factory default	value scope
M3-3-1-0	Up level	up-leveling parameters	000	0—500
	Down level	down-leveling parameters	000	0—500

M3-3-5 =Load set starting torque parameters

key value	display content	description	factory default	value scope
M3-3-5	speed low sect.	Elevator speed is less than the percentage for the speed of the rated speed section 1	0	0-100, choose 0 to cancel the additional speed adjustment
	speed mid sect.	Elevator speed is greater than the speed of paragraph 1 and the percentage is less than the rated speed for speed paragraph 2, greater than the speed of 3 for speed of 2 paragraphs	0	0-100
	speed Kp1	Speed 1 vibration suppression ratio adjustment coefficient	10	10-10000
	speed Ki1	Speed 1 vibration suppression of integral adjustment coefficient	10	10-10000
	speed Kp2	Speed 2 vibration suppression ratio adjustment coefficient	10	10-10000
	speed Ki2	Speed 2 vibration suppression of integral adjustment coefficient	10	10-10000
	speed Kp3	Speed segment 3 vibration suppression	1000	10-10000

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		ratio adjustment coefficient		
	speed Ki3	Speed segment 3 vibration suppression integral adjustment coefficient	1000	10-10000
	speed fliter	Speed loop filter coefficients, the larger the value of filtering the weak	40	1-1000
	Pretorque delay	Adjust the brake command to start and torque calculation time delay	800	0-1000
	Start inner Kp	Torque adjusting proportion coefficient of 2	0	0-10000
	Start inner Ki	Torque adjusting integral factor 2	0	0-10000
	Run source SVT=0	Command source	1	0, 1
	Max Drive ErrorB	Class B maximum allowable failure time	05	0—15
	Max Over Current	Over current maximum allowable failure time	03	0—5
	Start Kp	Start the proportional controller gain	2000	0 ~ 9999 note: asynchronous machine is set to 0
M3-3-5	Start Ki	Start the differential controller gain	200	0 ~ 9999 note: asynchronous machine is set to 0
	Vs abnorm T 0.1s	V yards down time	10	0—45
	NTSD PRESENT %	Forced to slow movement speed	097	0—200
	DDP time (0.1s)	Leveling photoelectric protection time	40	0—45
	Distance Comp	distance compete	03	0—99
	Max fdbk error	Current feedback maximum allowable failure time	3	0—5

Note: the speed loop parameters

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M3—3—1 (code menu)

Key	Display content	explain	Factory default	Value scope
M3—3—1 (S7-S4-S1-S2)	Drive size	Driver size	203	201~209
M3—3—1 (S3-S6-S9-S8)	Switch frequency	Switch frequency	10	6—12
M3—3—1 (S4-S6-S1-S5)	Ld0	Daxis inductance of 1.8 times	30	1—99
	Lq0	Qaxis inductance of 1.8 times	30	1—99
	Ld(mh),Lq(mh)	Daxis nductance, namely main engine inductance	30	1—99
	Lq(mh)	Qaxis inductance, namely main engine inductance	30	1—99
	Mutual resist	stator resistance	4	According to main engine
	Control methord	Main engine control mode	3	3 is synchronous machine , 1 is asynchronous machine

The host inductance parameters reference:

rating of machine	Ld (mH)	Lq (mH)	Lq0 (mH)	Ld0 (mH)
340V,26A,167,12P	8	8	13	13
340V,26A,209RPM,12P	8	8	13	13
340V,11A,95,12P	6	6	10	10

Note: if the host name plate did not mark the inductance value, you can refer to this table. Generally, the increase of rated current, inductance value decreased.

Part 5. Driver debugging

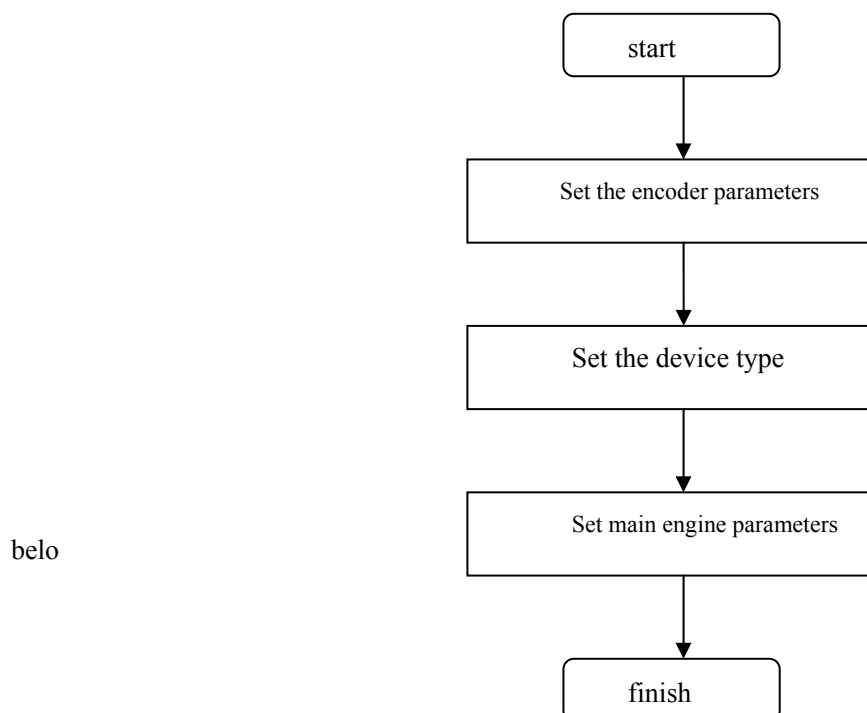
5.1 set running state

M3319 Run enable	M335 Run source	Enabled state
0	0	Initialization parameters, manual operation
0	1	Hoistway self-learning and

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		maintenance operation
1	1	inspection travel ,express run,reset run

For synchronous motor, the driving part before debugging, should press overhaul upward or downward button, to host a static automatic positioning. Drive part of the overall debugging process can see



5.2.1 static synchronous host encoder automatic positioning

1, 3-5 menu will M3 - RUN SOURCE SVT is set to 1.

2, input parameters: the correct M3-3-1-1 encoder parameters, and will be 4615 M3-3-1 password menu Control method is set to 3, 7412 M3-3-1 password menu at the same time in the Drive size is set to the corresponding device type, M3 Ld0 in 4615-3-1 password menu, Lq0 for 13, Ld, Lq of 8, and M3-4-3-1 the host parameters

3, the Encoder Dir set to 0, the monitoring server 3-1-1 Output current, hold the room overhaul upward or downward button, drive for automatic positioning, positioning process for 3 to 5 seconds (positioning need to hold in the process of overhauling upward or downward button), the current host hum the sound, the Output current value should be within 10% of the rated current. Positioning process after the motor running, if the motor running smooth, speed feedback is correct, no load Output current should be under 1 a. If motor jitter, appear the current

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FDBK error failure, you need to change the sale of any two phase, after relocation if running normally, observe the running direction, if contrary to the actual change the direction of rotation 3-3-1 rotate dir.Runtime check up the 3-1-3 Now Position monitoring system normal value increases.

5.2.2 inverter jumper wire and the encoder connection checks

Synchronous motor is Germany HEIDENHAIN encoder (cosine rotary encoder, with the reference zero and two are cosine absolute position sensor).For different encoder, power source jumper HAMCB board of encoder J7 set is not the same, can choose according to encoder working voltage 5 v or 8 v.HAMCB board jumpers and encoder connecting cable in the factory is ready, if discover the server according to encoder failure or motor can not run normally, in the case of ensure parameter is set correctly, please check these two items.**Encoder cable connection shown in the table below:**

The encoder interface	Wire color	interface	External PIN
1a	brown(C-)	Sin- Abs	11
6a	orange(D-)	Cos- Abs	13
2a	grey(A-)	Sin- Inc	5
5a	white(B-)	Cos- Inc	1
7b	purple(C+)	Sin+ Abs	10
2b	Pink(D+)	Cos+ Abs	12
6b	Greenish blue(A+)	Sin+ Inc	6
3b	green(B+)	Cos+ Inc	8
1b	red(+5V)	+5V	9
5b	black(0V)	0V	7
4a	Silver grey(R-)	Zero-	4
4b	yellow(R+)	Zero+	3

5.2.3 requires debugging when using asynchronous motor

Asynchronous motor with gear configuration is generally normal incremental encoder, pulse number 1024, the motor does not need to self learning, confirm the motor parameters are correct, and will be 4615 M3-3-1 password menu Control mothed is set to 1, while Inertia M3-3-1-1 kg/m2 parameter is set to between 1-10, 4615 M3-3-1 password menu Ld0, Lq0, Ld, Lq, four inductance values are set below 2, START in M3-3-5 Kp, START Ki are set to 0.J7 voltage jump line chosen to 8 v, J11, short J12 on the right side of the two pin, select incremental

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encoder, can be directly into trial operation;Encoder cable attachment please see electrical schematic diagram.

1, involves specific elevator parameters, such as deceleration parameters should be adjusted according to the actual situation of the elevator.Description: look from the traction wheel (stator) axial, traction wheel clockwise corresponding capsules upside, if the wrong, you can change the 3-3-1 rotate dir parameters, will run a reverse direction.

2, Gearbox thewire reduction ratio

The parameters for the gear ratio of the host.For gearless host, set this parameter to 1 (reduction ratio = 1).

3, the rope thewire rope

The parameters for the host than the product of the gear ratio and the rope.For gearless host, set this parameter to 2 (rope ratio = 2).

4, Pulley diameter

Tractive sheave diameter, according to the host tractive sheave diameter setting.5、 Duty speed

The rated speed, the value shall comply with the following equation.

$$\frac{FSS \times \pi \times PD}{60 \times GR \times RR} = V$$

FSS: Duty speed motor speed (RPM)

PD: Pulley diameter error!Reference source was not found.(m)

GR: Gearbox thewire error!Reference source was not found.

RR: Rope thewire Rope

V: Duty line speed for lift elevator speed (m/s)

错误！未找到引用源。

6 check elevator running direction

Into the maintenance operation mode

Inching runtime, need to confirm the direction of the lift is the same, if not consistent, need to adjust.

Run at the elevator, the elevator running direction and instruction direction are consistent

If inching upward, and the actual elevator is running down (or vice versa), elevator running direction and instruction in the opposite direction, the HAMCB plate driven part of the menu, the M - 3-3-1-1 will Rotate dir parameter inverse.

Save parameters, after confirm the elevator operation direction and the direction of the instruction is consistent.

7 Inching operation mode

7.1 the emergency electric operation (ERO)

Confirm the cabinet that the emergency electric operation switch (ERO) position as maintenance, car repair switch (TCI) in normal position.

With tci server input (M - 1-1-2) observation, uib, dib, the condition of ero, at this time on the server should

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display "tci uib dib ero".

Inching upward from the box of ERO button, display "uib" on the server strain uib "capital".

Point move ERO downward button on the box, display on the server to uppercase "dib dib" strain ".

Continue to hold down the up button, confirm the elevator running up.

Continue to hold down the button, confirm the elevator running down.

7.2 the car roof repair operation (TCI)

Will the car roof maintenance position switch (TCI) to repair, machine control cabinet emergency electric switch (ERO) to normal position.(note: the car roof repair switch operation, must comply with the "into its top operating procedures", the protection logic control part, the elevator will not be able to run, the server will be flashing "TCI - LOCK" of information, at the same time logic control part of the electronic board on the INS will flashing lights.)

At this time on the server tci into capital tci.

At the same time hold the overhaul box upside button U and C button, confirm the elevator running up.

Downward button and hold overhaul box of D and C button, confirm the elevator running down.

Carefully make the elevator to overhaul state run inside the shaft, ensure no outstanding obstacles blocking the elevator in the shaft running.If you have, take corresponding measures.

At the top of the car and checked by TES (its top urgent stop switch), EEC (safety switch) window, SOS (safety clamp switch), and the function of the upper and lower limit switch is valid.**8 position reference system adjustment**

8.1 the adjustment of the limit switch

Adjust the distance of the limit switch according to the following table (the distance of permissible error should not exceed 20 mm).Value plus or minus in front of it is for sure: in order to make the terminal floor elevator in the upper and lower layer positioned as a benchmark, in the guide for a token 00 mm.

For the attic room, plus said on the mark, minus said under this mark.

On the ground floor, plus said under this mark, minus sign said on the mark.

speed(m/s)	1LS,2LS	7LS,8LS
0.4	-80(nts:1000)	150±50
0.5	-100(nts:1300)	150±50
0.75	-220(NTSD:1300)	150±50
1.0	-390(NTSD:1300)	150±50
1.50	-870(NTSD:1300)	150±50
1.75	-1200(NTSD:1300)	150±50
2.00	-1550(NTSD:1300)	150±50
2.500	-2450(NTSD:1300)	150±50
3.000	-3500(NTSD:1300)	150±50

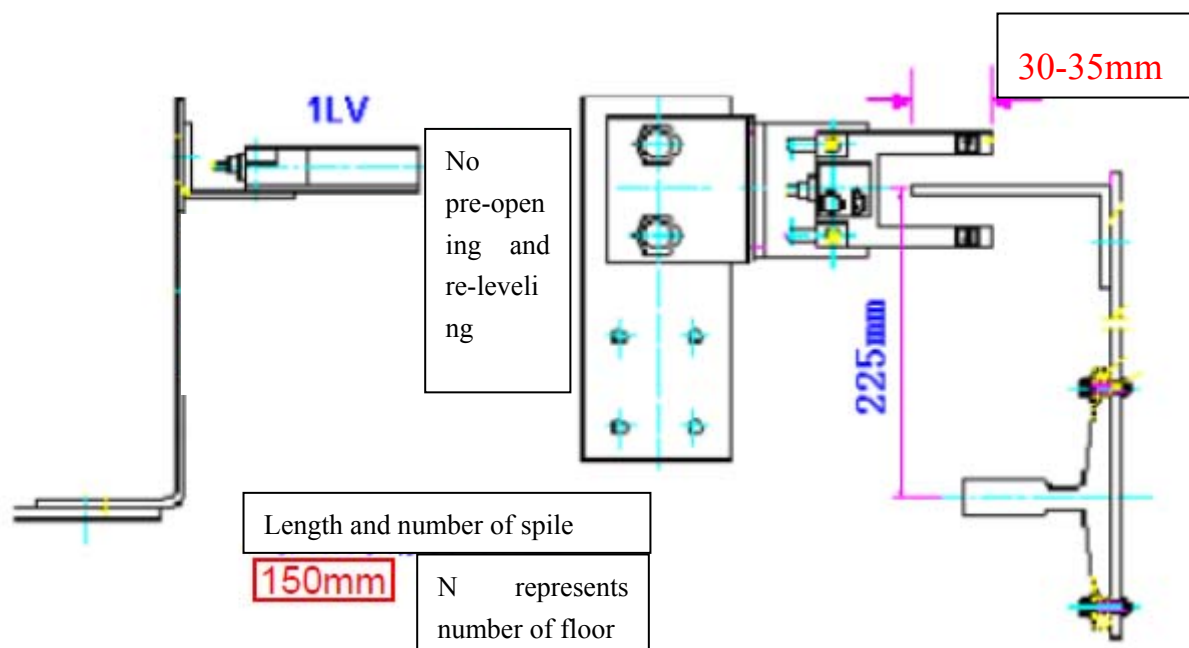
Remarks: The distance refers to the distance limit switch contact opened, rather than the distance limit switch when the roller pin connecting rod

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Adjust car roof photoelectric switch & hoistway light partition board

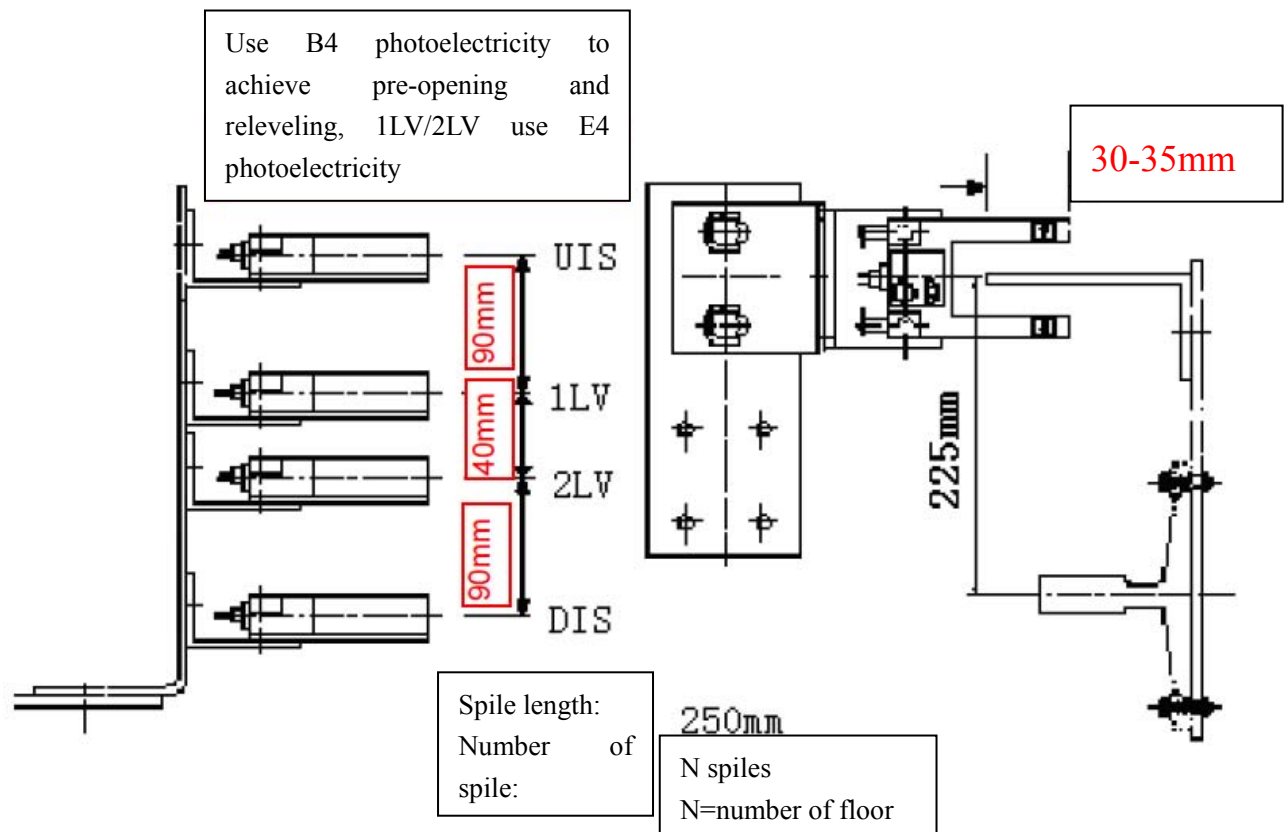
The installation method and dimesions of car roof photoelectric switch & light partition board as bellows:

No advance door opening and releveled function using Omron E4 singel photoelectric, the leveling board length 150mm



Advance door opening and releveled function using 4 Omron photoelectrics, UIS/DIS: B4 photoelectric, 1LV/2LV: E4 photoelectric, installation dimesions as bellows:

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Put balanced load in car(about 45% of the load)

Operate the elevator to the leveling position of each floor

Adjust light partition board of each floor, make its centerline is same as the centerline of two photoelectrics (means the centerline of 1LV&2LV is 30mm), this operation will effect the elevator leveling precision

9 First normal operation preparation

9.1 Safety, door lock loop check

Confirm each safety switch of safety loop work, under the condition of maintenance operation can stop the elevator when any safety switch is opened (OS,8LS,7LS,SOS,TES, PES, GS, DS, GSS) (Remarks: Check whether hoistway door lock of every floor work)

9.2 Check the signal of hoistway

ERO or TCI used in the whole operating elevator, check door area 1LV、2LV, forced to slow down 1LS、2LS etc's hoistway signal,; enter SVT1 by service, enter M1-1-2 and check input signal of HAMCB panel

10 self study of hoistway location

10.1 Before self study of hoist position, using ERO during the whole operating, using server to check and confirm the hoistway photoelectrics and forced speed-reducing switch work

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10.2 When the elevator is in the leveling, DZ、1LV、2LV should be capital letter; when running down the following photoelectrics work first, because that 2LV and DZ are converted to lowercase letters, upward running is on the contrary

10.3 When the elevator running around the bottom, 1LS work, so 1LS are capital letters; when the elevator running to the top floor, 2LS work, so 2LS are capital letters; when the elevator running to the middle floor, 1LS、2LS should be converted to lowercase letters

10.4 Confirm the floor number of M-3-3-1-9, HAMCB drive section, and set the parameter of RUN ENABLE to 0, the RUN SOURCE SVT of M3-3-5 to be 1, and confirm the TOP data of M1-3-1-1.

10.5 Turn the ERO and TCI to the normal position, operate drive section M-3-3-4 by server, press Shift&Enter to start self learning hoistway

10.6 The elevator will be running to the bottom at the low-speed, then do self learning by rising up at the self-learning speed, after reached the top floor, self learning finished.

10.7 After self-learning change the parameter of RUN ENABLE of M-3-3-1-9 to 1, the information of hoistway location is in the menu of M-3-1-4, please check again,

10.8 Dail ERO to maintenance location, running by press downward button to make the car stop at the non-levelling location, then dail ERO to right location, now the elevator should be reset operat, monitoring M-1-1-1 by server, displays COR, until reset to the bottom, now the M1-1-1 display CHC.

11 Normal operations

11.1 According to the wiring diagram to check the address of remote station RS5 and pin connector

11.2 Connect the server to SVT server interface of HAMCB board, according to the parameter list and I/O port of input&output table, to confirm all parameters and input&output address of the EEPROM is correct

11.3 Input the calling signal by server

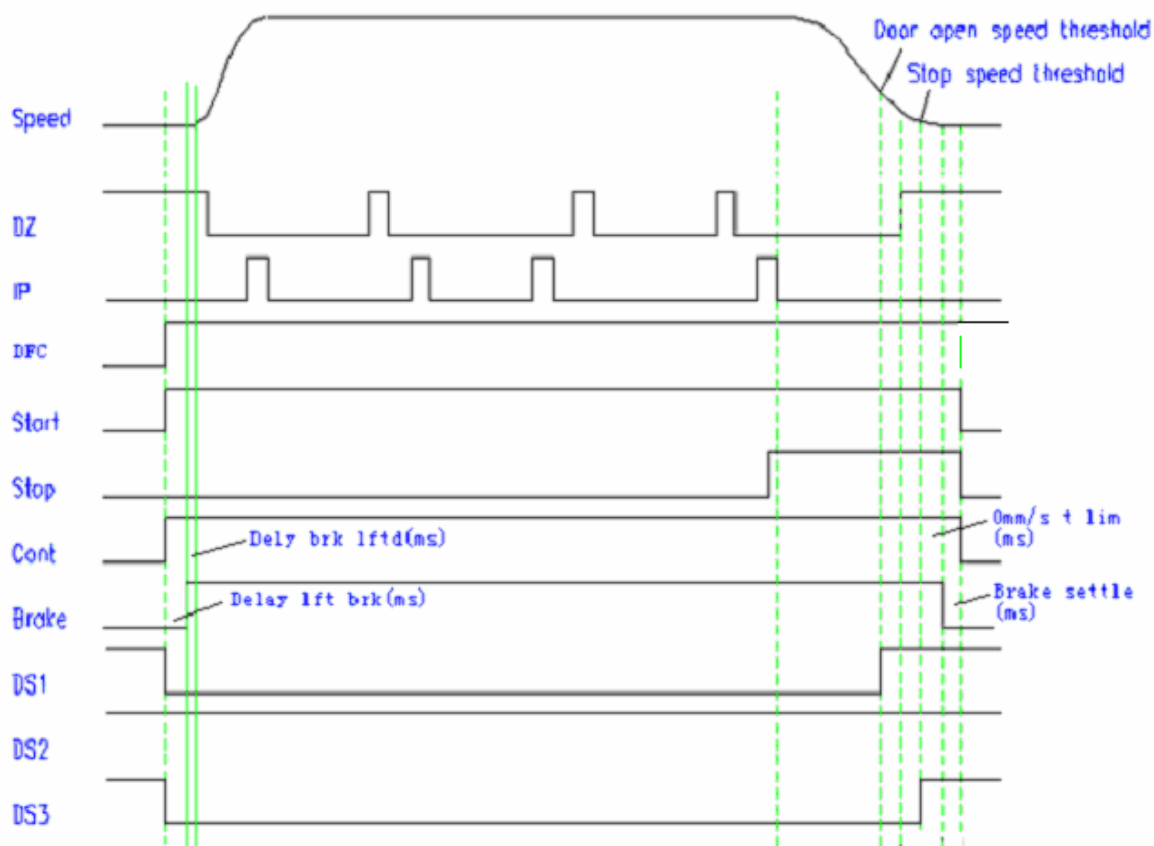
11.4 Press M-1-1-1 of server in turn, then input the calling signal and press the number of the call floor(Floor 1 corresponding to Key 0, Floor 2 corresponding to Key 1, etc)

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11.5 Press blue Key (shift), then press Key “ENTER”, the elevator should run into the call floor

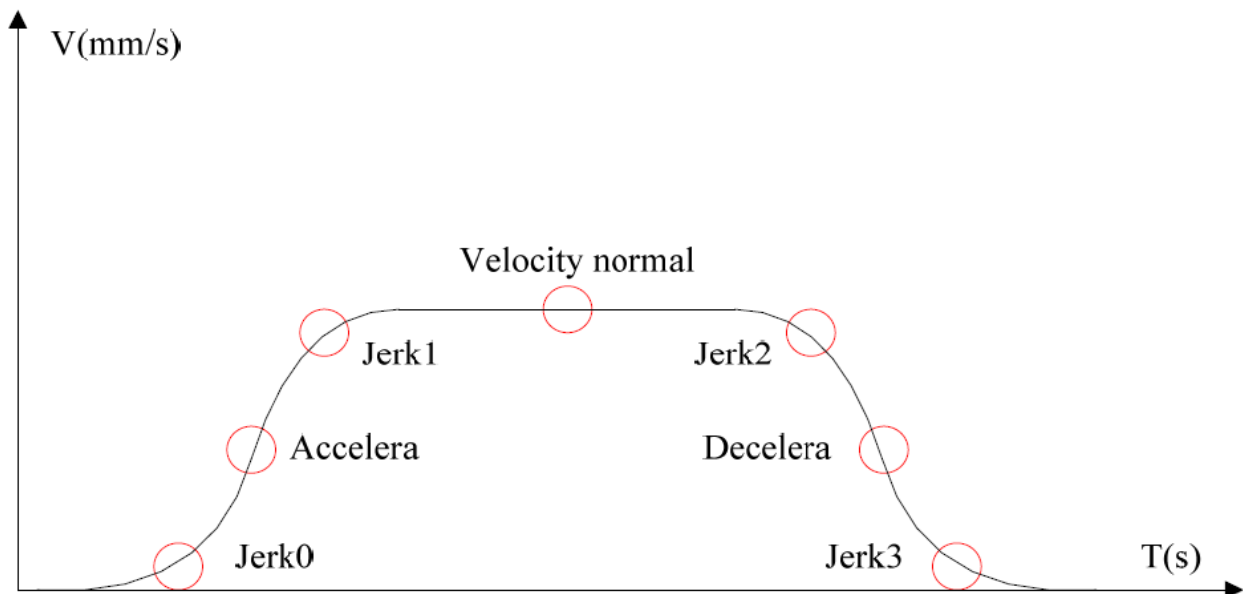
11.6 Check and confirm that all functions related to RS5 and RS32 are normal

11.7 Monitoring the Car speed of M-3-1-1 by server, during the whole operation of elevator to monitor whether the elevator to run at the contract speed



Elevator Normal Running Sequence Diagram

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Elevator Normal Running Graph

12 Adjust the flat position of normal running

12.1 Before electrical adjust the flat, should confirm finished the adjustment of the mechanical head area spile

12.2 Running down layer by layer, record the flat deviation of each floor, according to the average of deviation to amend DOWN LEVELC number of M-3-3-1-0

12.3 Running down layer by layer, record the flat deviation of each floor, according to the average of deviation to amend UP LEVEL number of M-3-3-1-0 (if deviation is large, adjust data)

12.4 Finish the adjustment of leveling position

13 start comfort adjustment

Put the elevator in CHC, DDO state, run elevator at the high speed to observe the traction wheel when it's start whether jitter sliding or not

If it's jitter, the parameter matching performance is not good, enter M-3-3-1-1 first to adjust system inertia

Increase or decrease 5 every time, Generally if the system inertia is too high, when the elevator stopped will high frequency jitter, if the system inertia is too small when the elevator start will be power shortage and sliding

If after adjust system inertia, it's still sliding, then need to adjust M-3-3-5 "Start Kp" and "Start Ki" at the same time, generally if the system is strong rigidity(no car roof spring), set up Kp3000-7000,Ki150-700. If there

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is car roof spring, set up KP1000-3500,KI100-350. There is best associated value within this range for KP,KI.

If it's still sliding after adjusted Kip,Ki, adjust M3-3-1-6 APR time when traction machine open delay, APR time provide the time of duration for output starting torque,there is best associated value between 100 to 400 for APR time.

Adjust the parameters as above, observing elevator, until achieve the best state for starting

14 operate comfort adjustment

Put the elevator in CHC, DDO state, run elevator at the high speed to observe the traction wheel running from low speed to high speed whether jitter sliding or not

If it's jitter, start speed of segmentation adjustment, enter M-3-3-5 to adjust the value of Speed Low Sect and Speed Mid Sect.

Set up three speed segments, less than the percentage of rated speed of Speed Low Sect is low speed section(Speed One), between the percentage of rated speed of speed low sect and speed mid sect is Speed Middle Sect is low speed section(Speed Two), more than the percentage of rated speed of Speed Mid Sect is high speed section(Speed Three)

Such as adjust vibration on the high speed section(Speed 3), need to adjust Speed KP3 and Ki3 at the same time.

There is best associated value within adjustment range for Speed KP3, KI3(factory defaults 1000,adjustment range 10-10000), when adjusting, increase or decrease synchronously, the adjustment value is 200 every time; If the traction machine apparent abnormal jitter in the process of adjusting, should be appropriate synchronization decrease this parameter, until the optimal operation.

15 ARD function(Automatic emergency evacuation device)adjustment instruction

When the elevator is running, if the system power supply suddenly loses power, passenger may lock in the car. For this kind of situation, the integration of system design a blackout emergency operation scheme. UPS provide the power supply for the main loop of the system and the working power, to blackout emergency operation

After system in ARD mode, running at the speed of M-3-3-1-2 ARD Speed(mm/s), Direction is elevator light load direction, if running direction for the first time is wrong, change the parameter of ARD Run Direct, to set up direction(factory faults 0)

When detected the flat layer signal, keep the door open, the elevator stop running.

3 segments of ARD operation, as bellow

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Server display	f u l l n a m e	explanation	section
EPC	Emergency Power Control	Emergency power control	after received power grid signal then emergency standby power to rescue operation to be performed
EPR	Emergency Power Rescue	Emergency standby power to rescue	Perform emergency supply rescue operation
EPW	Emergency Power Wait	Emergency standby power to wait	ARD completed, running at the flat layer and keep the door fully opened, to enter the state of emergency standby power waiting

I/O related parameters setting

board	I/O	Default value	Preset value
HAMC B	17(generator signal)	Logic Add=0 Add=00 A=0	Logic Add=0 Add=01 A=0
	1676(virtual flat signal)	Logic Add=0 Add=00 A=0	Logic Add=0 Add=63 A=1
	1677(back door signal)	Logic Add=0 Add=00 A=0	Logic Add=0 Add=63 A=2

Related parameters setting

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Menu	Parameter name	Factory fault	Function and range
M3312	ARD-Speed (mm/s ²)	160	Function: rescue operation speed of emergency prepared power and the flat layer precision Parameter setting range: 1-500; based on the precision requirement of flat layer, suggest setting range 80-150
M3312	ARD-RUN DIRECT	0	Function: depend on the motor installation left or right Parameter setting range: 0-1 Motor left installation=1; motor right installation=1
M18-8	EPO-DC	0	Function: after ARD finished rescue, keep door open Parameter setting range: 0-255(s)
M1315	EPO-P	255	Function: EPO stop position When start integration ARD function, the parameter must be 64
M18-10	EN-EVT	0	Function: whether to save the logical events after motherboard power off 0: not be saved 1:saved When start integration ARD function, the parameters must be 1

ARD function testing process

After finished all connection and parameters setting, test ARD function, work procedure as follows:

1、 Confirm the elevator system is in normal operation model

2、 Cut off the main power of control box, but not the air switch of the control box

3、 Monitoring system of server M-1-1-1 is in EPC state

4、 When EPC model to EPR model, elevator start rescue operation

5、 When the elevator is in ARD model, it will running in the direction of light load. For example, when the elevator is in no-load state, car running up; On the contrary, when the elevator is full-load, car running down. The elevator doors open to release the passengers when running at certain floor, t

6、 When car stop at door open, car door opened, the monitoring system of server M1-1-1 is in EPW state. According to change the parameters of EPO-DC to change the waiting time of opening door

7、 When recover the main power of control box, the elevator will be turned into the normal model

16 real time clock function explanation

System integrates the function of real time clock; customers can set the clock with the standard time

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synchronization, according to the need to set the timing function. For example, you can set the once upon a certain time when elevator stopped.

There are two setting interface for standard timing setting and clock controlled setting

Standard

clock setting

M1-6 standard clock interface



Time setting by press SHIFT and 1



- 1: MON
2: TUE
3: WED
4: THU
5: FRI
6: SAT
7: SUN

Input the right time and date then press Enter

The clock control set

M1-3-1-6 Clock control interface

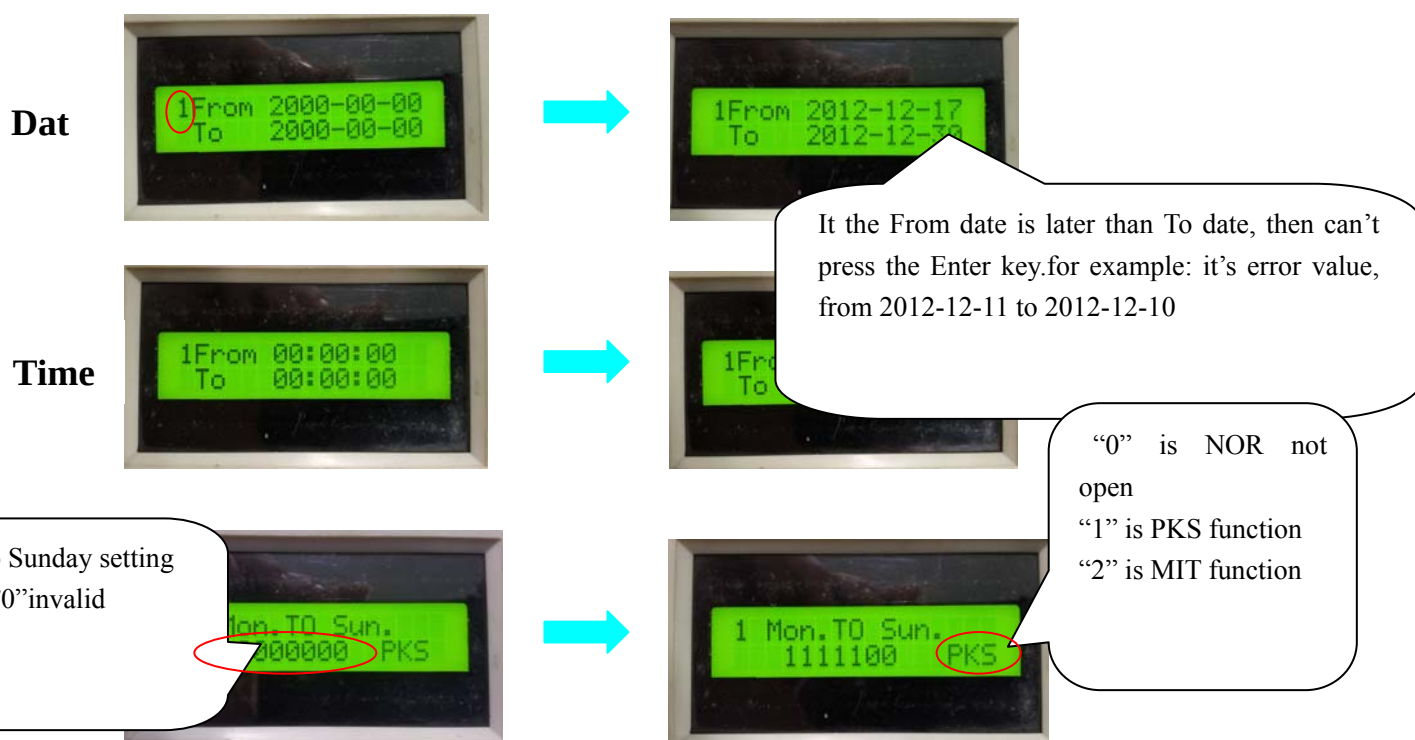
set up functions of the first set

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3 set of timing setting, “1”means first set,

If the number is higher than the actual value, then can’t press the Center key.for example: 2012-13-10 is error value

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The setting above: from 2012-12-17 to 2012-12-30, every Monday to Sunday, from 8:30 to 14:30 open elevator lock function, other time normal operation

When according to the time cycle to start the elevator lock function, dial the lock switch to locked state, then dial to Normal, after that "locked" cancelled. Until start the next clock locked, then locked again.



The third set of the clock function is custom clock; customer can set up output IO 1780.

Clock function group 4 and 5 is customized clock, customer can set time output, IO not output. For example, set MUTE mode, can control IO1092, IO1093 to realize MOTE mode

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17. Logic functions and drive parameter table

17.1 The logical parameters and Settings

17.1.1 Parameter table 17.1.1 logic

M1-3-1-1

No.	symbol	default	range	meaning	remark
0	TOP	7	1~63	attic room	(contract parameters)
1	LOBBY	0	0~63	base station	(contract parameters)
2	BOTTOM	0	0~63	Ground floor	
3	CFT-P	255	0~255	Restaurant layer: > 31 is invalid	
4	CFT-NT	255	0~255	Restaurant layer door holding time: If MIN - H is beyond time, this time will be used on the CFT is - P. 0 (0.1) 0.1 s	
5	ARD-P	255	0~255	The base station set up If there are no other outbound, capsules set after ARD - T time to return to the floor 0 (1) 31 > 31 is invalid Note: only single ladder	
6	ARD-T	0	0~255	ARD (to be automatic return to the base station) mode delay Settings 0 (10) 2550 s	
7	PKS-P	0	0~255	Ladder lock service location choice When PKS operation, capsules moved to the specified location, open closed the elevator after a shutdown, only open button is valid 0 (1) 31 > 31 is invalid	
8	PKS-TY	0	0~2	Lock the ladder type: 0 after completion of all call service elevator back to the parking position after PKS - T time to stop 1 to 0, but the elevator stopped after the switch to the ISC 2 to 0, but the elevator door open after stop	

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9	PKS-T	5	0~255	Ladder lock time In the PKS - P capsules after downtime O (1) 255 s	
10	EN-IST	0	0~7	Allows safety inspection service IST (intermittent stop) : Stop at IST -p if the car passes this landing. 0 is prohibited 1 the direction only 2 only under the direction of 3 the two sides to 5 if ISTS start, only on the direction 6 if ISTS started, only under the direction 7 ISTS started, only under the direction	
11	IST-P	255	0~255	The security check stop position: O (1) 31 > 31 is invalid	
12	EN_CRT	2	0~1	Use the contract processing: 0 up to 16 layers 1 up to 32 layers 2 up to 48 layers 3 up to 64 layers	
13	CR-DAR-T	15	0~30	The rescue operation, open the door waiting time 0 no open the door waiting time 15 relief when I opened the door waiting for 15 seconds	
14	CR-OPT	0	0~15	Rescue mode choice:0 no open rescue function; 1 start the rescue operation, when there is no choice + 8, said in a floor with hall door installation APRS;Dock 2 rescue the gate position4 when rescue floor opens the door, cancel the buzzerIn the absence of hall door floor installation APRS;Set to 1 3 said (1 + 4 + 8) open reduction and relief layer function, cancel the rescue flat layer to open the door to the buzzer, and in the absence of hall door installation APRS position	

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15	ATT	0	0~8	The driver mode selection:0 hold most of the latter RDCB until the door shut down completely 1 hold ATTU or ATTD until the door shut down completely 2 to 0, without calling the door closed0 + 3 function button in the car1 + 4 function button in the car 5 function 2 + button in the car0 + 6 function button in the car + buzzer1 + 7 function button in the car + buzzer2 + 8 function button in the car + buzzer	
16	ISC	1	0~9	Independent mode selection:0 for CCB button 1 last DCB/RDCB button 2 the use of ISU/ISD button control box 3 equal to CHCS (commissioner called), and continue to press the DCB/RDCB 4 moving CCB0 + 5 function independent of VIP model, hall outside not display1 + 6 function independent of VIP model, hall outside not display2 + 7 function independent of VIP model, hall outside not display3 + 8 function independent of VIP model, hall outside not display4 + 9 function independent of VIP model, hall outside not display	
M1-3-1-2					
No.	symbol	default	range	meaning	remark
0	DRIVE	0	0~1	Drive type: 0 no open early function 1 open early function	
1	EN-RLV	0	0~1	Releveling operation Settings 0 is invalid 1 start	

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2	HL-SET	0	0~2	Landing the lantern/bell set 0 landing the lantern and the bell with the same output 1 landing the lantern and bell with different outlet 2 use in box	
3	CR-DIR	0	0~1	Correction operation preferred direction: 0 down 1 up	
4	EN-SHB	0	0~1	Independent button to start the capsules 0 outside the normal operation 1 after the door shout independent operation	
5	SPEECH	1	0~1	Voice mode: 0 only capsules floor information from time to tome 1 capsules, and to call a floor information from time to tome .	
6	GCBTYP	0	0~2	The button control type: 0 is prohibited GCB - 1 (ban capsules instruction; allow parking operation) 2 GCB - 2 (banned all instruction)	
7	EN-CRO	0	0~7	Reader: 0 is invalid 1 signal contact card reader Using CRC contact and Card - Rd Mask, C bit = 1: always allow capsules instructions C bit = 0: capsules instructions only allowed if the CRC operation 2 combined contact card reader Using CRSn contact and Card - Rd Mask, C bit = 1: always allow capsules instructions C bit = 0: capsules instructions only allow corresponding CRSn contact operations More than three signals and contact card reader 3In combination with 1 and 2, so the two CRC and CRSn can be used for the same time 4 no card reader	

				5 same as 1, but the ISC will cover any CRS safety 6 same as 2, but the ISC will override any CRS safety 7 same as 3, but the ISC will cover any CRS safety	
8	DZ-TYP	1	0~2	DZ code: 0 DZ = 1 lv 1 DZ = 1 lv 1 and 2 lv 2 POSY operation (if there is no one nor ADO RLV) Parking: DZ = 1 lv lv and 2 After parking: DZ lv lv or 2 = 1	
9	DZ-DLY	250	0~255	DZ delay: 0 (0.004) 1 s (LV-MOD = 0)	
10	RLV-TY	0	0~1	Releveling type RLV: 0 RLV releveling 1 DRLV delay releveling(用 DIS1)	
11	RL-CNT	3	0~255	Releveling steps limits RLV: 0 (1) 255	
12	RL-DIS	255	0~255	Set in flat layer up time (DIS lost): 0 (0.1) 25.5 s	
13	RL-UIS	255	0~255	Set in flat layer down time (UIS lost) : 0 (0.1) 25.5 s 0 (0.004) 1 s (hydraulic drive)	
14	RL-U-D	0	0~255	Set in flat layer delay up: Used for precision in flat layer stopped during the period of adjustment 0 (0.004) 1 s	
7	EN-CRO	0	0~7	card reader: 0 is invalid 1 signal contacts card reader Use CRC and Card-Rd Mask, C bit = 1: cabin indication is always enabled C bit = 0: cabin indication is enabled only if CRC operates 2 combination contact card reader Use CRSn and Card-Rd Mask, C bit = 1: cabin indication is always enabled	

				C bit = 0: CRSn 触点操作 cabin indication is enabled only on the condition of relevant CRSn contact operation 3 signal and multiple-contact card reader Combine 1 and 2, so CRC and CRSn can be used at the same time 4 no card reader operation 5 is same as 1, however ISC will cover all security of CRS 6 is same as 2, however ISC will cover all security of CRS 7 is same as 3, however ISC will cover all security of CRS	
8	DZ-TYP	1	0~2	DZ 码 code: 0 DZ = 1LV 1 DZ = 1LV and 2LV 2, POSY operation (If no RLV, no ADO either) stop: DZ = 1LV and 2LV after stopping: DZ = 1LV or 2LV	
9	DZ-DLY	250	0~255	DZ delays: 0 (0.004) 1 s (LV-MOD = 0)	
10	RLV-TY	0	0~1	RLV: 0 RLVreleveling 1 DRLV delayed releveling (use DIS1)	
11	RL-CNT	3	0~255	limit for releveling step RLV: 0 (1) 255	
12	RL-DIS	255	0~255	Set in the time that leveling ups (DIS lost): 0 (0.1) 25.5 s	
13	RL-UIS	255	0~255	Set in the time that leveling downs(UIS lost): 0 (0.1) 25.5 s 0 (0.004) 1 s (hydraulic drive)	
14	RL-U-D	0	0~255	Set when leveling delay up: Used for precision when leveling shuts off during the adjustment period 0 (0.004) 1 s	
15	RL-D-D	0	0~255	Set when leveling delay down Used for precision when leveling shuts off	

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				during the adjustment period 0 (0.004) 1 s	
16	PI-POS	0	0~3	Leveling change Settings: 0 display of leveling changing based on IP 1 display of leveling changing based on IP and DZ 2 when leave the leveling, floor display shows the change 3 when come to the leveling, floor display shows the change	
17	OLD-TYP	0	0~1	Overload mode 0 OLD mode, calls from COP are all cancelled 1 OLD mode, calls from COP are not cancelled, and can not register any other calls	
18	INS-TYP	0	0~1	Maintenance no- disturbing mode: 0 Close maintenance no- disturbing mode 1 Open maintenance no- disturbing mode	

M1-3-1-3					
No.	symbol	default	range	meaning	remark
0	DOOR	5	0~13	The front door type: 5 relay signal 12 encoding signal	
1	REAR	0	0~13	The back door type: 5 relay signal 12 encoding signal	
2	DBP-TY	1	0~1	Door branch circuit type: 0 DBP operation (4 - relay) MCS310 MCS310 1 LVC run (3 - relay) MCS120(M), MCS220(M)	
3	LR-T	240	0~255	Lighting delay time (IO 0026) : When car stops, after LR – T delaying, car lights and ELD will shut down	

				If the LR - MODE = 0 or 1, the setting range is 0 to 255 seconds If LR - MODE = 2 or 3, the setting range is 0 to 255 minutes	
4	LR-MODE	0	0~1	Fan/light delay mode selection (IO 0026) : 0 LR only if the door is closed, will work after LR - T seconds (lighting) 1 LR even if the door is open, will also work after LR - T seconds (fan) 2 LR only if the door is closed, will work after LR - T minutes (lighting) 3 LR even if the door is open, will also work after LR - T minutes (fan)	
No.	symbol	default	range	meaning	remark
5	MAX-C	40	0~255	The car door time 2 0 (0.1) 25.5 s	
6	MIN-H	40	0~255	The Hall door time 1: 0 (0.1) 25.5 s	
7	MIN-C	20	0~255	The car door time 1: 0 (0.1) 25.5 s	
8	MAX-H	60	0~255	The Hall door time 2 0 (0.1) 25.5 s	
9	DCP-T	50	25~255	DCP mode delay time (car delay protection) : When opening time is longer than DCP - T, car will deviate from group operation.If EN – NDG works, door will close slowly again. 25 (1) 255 s	
10	DTC-T	20	10~255	Time for close protection: If the door can not fully closed, it will open again in DTC – T time, and then try to close.The elevator will lock after three times failure. 10 (1) 254 s DTC - time 255 is invalid	

11	DTO-T	20	10~255	Time to open the door to protect: If cannot open the door, under the DTO - T time goalkeeper try to shut down, and try to open again. But after three attempts failed elevator will lock 10 (1) 254 s DTO - time 255 is invalid	
12	LOB-NT	40	0~255	Main floor door: If the time is longer than MIN - H, it is for LOBBY 0 (0.1) 25.5 s	
13	DHB-T	120	0~1	Allow the validity of CARGO2000 elevator door hold button (IO 620628) DHB door time can be cancelled... 0 choose DCB or any others can cancel the door holding time 1 (at this moment the default door hold time DHB -t is 1 second) 1 choose DCB or any others, or choose DHB again to cancel door hold time (at this moment the default door hold time DHB -t is 10 seconds!)	
14	F:EN-DCL	1	0~1	Set the IO 694 to allow the DCL input 0 no DCL switch 1 set DCL of I/O694 Only applies to the general door machine operation. If the EN - DCL = 1 setting, though DCL will arrive soon, DC will be released, even if DC - TYP = 11 programming.	
15	F:DOL-D	0	0~255	Automatic simulation DOL 0 set DOL of I/O 000 0.1 (0.1) 25.5s if no DOL in stipulated time, then simulate DOL signal Note: only applies to the general door machine operation.	

16	R:EN-DCL	1	0~1	Set the IO 695 to allow the RDCL input 0 no RDCL switch 1 set I/O695 RDCL Only applies to the general door machine operation. If the EN - DCL = 1 setting, though DCL will arrive soon, DC will be released,, even if DC - TYP = 11 programming.	
17	R:DOL-D	0	0~255	Automatic simulation RDOL 0 set RDOL of I/O 544 0.1 (0.1) 25.5s if no RDOL in stipulated time, then simulate RDOL signal Note: only applies to the general door machine operation.	
18	EN-DDO	0	0~2	The door operation: Choose front and back doors open at the same time 0 This parameter is prohibited 3 Front and back doors response calls from car and hall at the same time 3 Front and back doors response calls from car, and response according to hall call	
19	NO. IN 1LS	1	1~2	1 Quantity of LS inner door area 1 one door area 2 two door areas	
20	NO. IN 2LS	1	1~2	2 Quantity of LS inner door area 1 one door area 2 two door areas	
21	CR-CHK-T	0	0~20	Hall door detection time for nearby leveling function 0 close hall door detection 5 start hall door detection of 0.5 seconds 20 start hall door detection for 2 seconds	
22	EN-CK	2	0~5	Used to cancel the door time CCB 0 is invalid 1 Use CK 2 Use CBC 3 Use CK and CBC 4 Use RCBC 5 Use CK and RCBC Note: CK: choose car call to cancel the door time	



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				CBC: cancel registered car calls (press twice when door is opening) RCBC = cancel registered car calls (press twice when elevator stops or is running)	
--	--	--	--	--	--

M1-3-1-4

No.	symbol	default	range	meaning	remark
0	GRP-NO	1	1~4	Group control number 1 (1) 3	
1	GROUP	1	1~4	Group units 2 (1) 3 Note: for CAC or lower version, four group control only applies to building which higher than 11 floors For DAA or higher version, four group control applies to LCB2 - ASIC or building which higher than 16 floors	
2	CNL	1	0~4	The quantity of the elevators in the hall: 0 prohibit group stop in the hall 1 (1) 3 The quantity of the elevators in the hall Note: set the CNL = 0 avoid stopping in the hall	
3	MIT-ST	0	0~255	Up- peak starts hold time MIT: If during the MIT – ST period, two cars are LNS load when leave the hall, then start MIT - ST	
4	MIT-T	0	0~255	The up-peak exit time: If during the MIT – ST period, two cars don't leave the hall or not LNS load, then the MIT - ST pauses	
5	MIT-VD	0	0~255	Time interval of dispatching on up-peak: If there is a call or someone into car, the specified car in the hall will leave	
6	MIT-DOOR	0	0~1	Door status of waiting elevator in the main floor on up-peak 0 door is opening waiting 1 door is closed waiting	
No.	symbol	default	range	meaning	remark

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7	MIT-NLB	0	0~1	Setting of back to the hall on up-peak 0 respond to the call when back to the hall 1 does not respond to the call when back to the hall	
8	DUPK-P	255	0~255	Boundary location of double group on up- peak: The elevator group is divided into two groups, if DUPK input is activated, then DUPK - P separates from the group. 0 (1) 31 > 31 is invalid	
9	DUPK-G	0	0~1	Group of double up-peak: 0 car belongs to group 1 Car services to hall floor, and all the floor under the DUPK - P, But doesn't respond to all calls 1 car belongs to group 2 Car services to hall floor, and all the floor under the DUPK – P, it responds to calls	
10	MOT-T	0	0~255	The down-peak exit time MOT: If in MOT time, two cars come to the hall with full load, then MOT starts. If only one car to the hall or two cars come to the hall with full load not at the same time, then MOT suspends to start	
11	EN-UCB	0	0~255	Allow no response to up calls 0 Cars respond to up or down hall calls 1 cars only respond to down hall calls	
12	TFS-P	255	0~255	convert The transfer layer location: If there is low-levelings and high-leveligns in the group, then packet transports passengers in this position. 0 (1) 31 > 31 is invalid	
13	PARK-1	255	0~255	Peak bell location 1: If the PCLK1 is acted, the car will be parked in the PARK - 1 instead of ARD - p. PCLK1 has the highest priority, while PCLK8 has the lowest 0 (1) 31 > 31 is invalid Note: only applies to single ladder	
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14	PARK-2	255	0~255	Peak bell location 2: If the PCLK2 is acted, the car will be parked in the PARK - 2 instead of ARD - p. PCLK1 has the highest priority, while PCLK8 has the lowest 0 (1) 31 > 31 is invalid Note: only applies to single ladder	
15	PARK-3	255	0~255	Peak bell location 3: If the PCLK3 is acted, the car will be parked in the PARK - 3 instead of ARD - p. PCLK1 has the highest priority, while PCLK8 has the lowest 0 (1) 31 > 31 is invalid Note: only applies to single ladder	
16	PARK-4	255	0~255	Peak bell location 4: If the PCLK4 is acted, the car will be parked in the PARK - 4 instead of ARD - p. PCLK1 has the highest priority, while PCLK8 has the lowest 0 (1) 31	
17	PARK-5	255	0~255	Peak bell location 5: If the PCLK5 is acted, the car will be parked in the PARK - 5 instead of ARD - p. PCLK1 has the highest priority, while PCLK8 has the lowest 0 (1) 31 > 31 is invalid Note: only applies to single ladder	
18	PARK-6	255	0~255	Peak bell location 6: If the PCLK6 is acted, the car will be parked in the PARK – 6 instead of ARD - p. PCLK1 has the highest priority, while PCLK8 has the lowest 0 (1) 31 > 31 is invalid Note: only applies to single ladder	
19	PARK-7	255	0~255	Peak bell location 7: If the PCLK7 is acted, the car will be parked in the PARK – 7 instead of ARD - p. PCLK1 has the highest priority, while PCLK8 has the lowest 0 (1) 31 > 31 is invalid	
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				Note: only applies to single ladder	
20	PARK-8	255	0~255	Peak bell location 8: If the PCLK8 is acted, the car will be parked in the PARK – 8 instead of ARD - p. PCLK1 has the highest priority, while PCLK8 has the lowest 0 (1) 31 > 31 is invalid Note: only applies to single ladder	
21	EN-ZBS	0	0~1	Allow separated basement area: 0 no idle elevator stops in the basement area 1 one idle elevator stops in the basement area	
22	PRKDST	0	0~31	Implement function of returning to the base station to the minimum distance between parking position and floor space Execution of parking operation, if the distance between leveling and destination is too long, this parameter provides the floor number. 0 (1) 31	
23	TPOS 1	2	0~255	Beginning floor 0 (1) 31 > 31 is invalid	
24	TPOS 2	3	0~255	The last floor: 0 (1) 31 > 31 is invalid	
25	TDELAY	0	0~255	Delay time 0 (1) 255 s	
26	NoAdrChk	1	0~1	Disable security inspection: Verify that the TCI (I/O 691) is programmed to effective RS address. 0 Using RSL safety address inspection 1 doesn't verify the address	
27	NoDW_Chk	1	0~1	Disable the DW security inspection during normal period: 0 Check the DW when open the door 1 doesn't check DW during normal period	
M1-3-1-5					
No.	symbol	default	range	meaning	remark
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0	EFO-P	0	0~255	The fire-fighting floor: 0 (1) 31 > 31 is invalid Note: this parameter is ignored EFO - NC = 1	
1	EFO-NC	0	0~255	The fire-fighting floor: 0 (1) 31 > 31 is invalid Note: this parameter is ignored EFO - NC = 1	
2	EFO-OP	0	0~3	The alternative function of fire-fighting EFO: Normal function of defective EFO input (such as smoke transducer) may be ignored, while EFO input will be restored 0 no alternative 1 Fire-fighting operation can be manually cancelled after returning to fire floor 2 fireman service instead of fire-fighting services 3 start the function of above 1 and 2 at the same time	
3	EFONDG	0	0~1	EFO allows that anti-pinch speed closed during fire-fighting 0 closed in maximum speed 1 anti-pinch speed close	
4	EFO-REV	0	0~2	EFO allows opening door again during fire-fighting service 0 all is forbidden 1 only SGS/DOS is allowed 2 DOB and SGS/DOS are allowed	
5	EFO-MP	0	0~7	The fire-fighting service priority mode EFO 1 ISC is prior (independent) 2 ATT is prior (driver) 4 EHS is prior (emergency medical service) Note: if you want to use the above function at the same time, you can sum the number (example: 1 and 4 service needs to input 1 + 4 = 5)	
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6	EFO-SD	1	0~255	Signal device status on state of emergency fire-fighting: 0 no buzzer 1 constantly flicker 2 always flashing 3 often flickers under not emergency power (EPO), while often flashes under emergency power	
7	ASL-P	255	0~255	Standby floor on condition of emergency fire-fighting service: If input standby floor, the car will move to ASL -p (standby floor)instead of EFO -p (fire-fighting floor) 0 (1) 31 > 31 is invalid	
8	EFS-TY	10	0~10	Fire control type: 0 EFO cars will move to fire floor, and stay there with opening door EFS 1 (Automatic) if the car arrives at the fire floor, it will automatically turn into the independent state. ISC. EFS 2 (Manual) if the car arrives at the fire floor, it can turn into the independent state by independent switch 3 ANSI (SES I / II) 4 The British standard 5533 5 Swiss standard 6 Australian standard Same as EFS 2, but cars can switch to the independent state (ISC) And ESK switch 7 New Zealand 8 EFS 2 (hand use ESK ESK) 9 New Zealand uses DCB 10 general	
9	EFSINI	1	0~31	Fireman starts standby power EFS 1 automatic 2 when operating independently ISS 4 when ESK starts ESK When CFS starts CFS 16 when 1EFS starts 1EFS Note: if you want to use the above function at the	

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				same time, you can sum the number (example: 1 + 4 = 5) Note: only applies to EFS - TY = 10	
10	EFSCLD	3	0~63	Door closed type of the fireman service EFS 1Calls from car (continuous pressing) 2 The closing button (continuous pressing) 4 Up/ down starting button 8 CS starting button 16 American independence 32 Calls from car(press gently) Note: if you want to use the above function at the same time, you can sum the number (example: 2 + 4 = 6) Note: only applies to EFS - TY = 10	
11	EFSOPD	2	0~15	Door opening type of the fireman service EFS 1 always automatically 2 Press DOB button continuously 4 only open the door at the first time 8 door holding function, if the car door does not open or close completely (I/Os DDM, RDDM) Note: if you want to use the above function at the same time, you can sum the number (example: 2 + 4 = 6) Note: the valid number 1, 2, 4 and 6 Note: only applies to EFS - TY = 10	
12	EFS-EX	0	0~3	Exiting type of the fireman service EFS 0 only in the fire floor (EFO) 1 no longer moving after door opening 2 after the door closed on any floor Note: only applies to EFS - TY = 10	
13	EFS-RC	0	0~63	Recall type of fireman service EFS 1 EFK closes timeout 2 EFK is closed, elevator is running 4 EFK is closed, the elevator is not running 8 EFK is closed, the elevator is not running, the door is closed 16 EFK is closed, the elevator is not running, the	

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				door is open 32 EFS1 cancel the fire-fighting Note: only applies to EFS - TY = 10	
14	EF-I-LT	0	0~15	Fire control input locked EFO/EFS 1 EFO input locked 2 AEFO input locked 4 EFK input locked 8 ASL input locked Note: if you want to use the above function can at the same time the numerical sum (example: 1 + 4 = 5) Note: only applies to EFS - TY = 10	
15	EPO-P	255	0~255	Emergency power position EPO (emergency power operation) : O (1) 31 EPO A/C (setting location) EPO E (hydraulic equipment: the ground floor) = 32 EPO B/D (the next floor) EPO F (hydraulic equipment: the next floor) > 32 is invalid	
16	EQO	0	0~1	Earthquake operating version: 0 California 1 New Zealand	
17	EPS-TYP	0	0~132	Fast priority services: The EHS instruction is given after arriving at the appointed floor: xx0 Automatical Independent ISC xx1 Return to normal xx2 Cancel hall calls automatically CHCS The EHS instruction is given before arriving at the appointed floor: x0x return immediately x1x stop in the next floor x2x Allow to register car calls during running x3x serve all car calls General: 1 xx Medical service lamp (EHS - 2)	
17.1.2 Elevator operation setting					
Allowed ENABLE Masks (SVT Key Sequence M-1-3-3-1-GOON)					Elevator
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operation setting					
At level	CUDE	CUDE	P	R	Remark
0	1100	0000	0	0	contract arameters
1	1110	0000	0	0	contract arameters
2	1110	0000	0	0	contract arameters
3	1110	0000	0	0	contract arameters
4	1110	0000	0	0	contract arameters
5	1110	0000	0	0	contract arameters
6	1110	0000	0	0	contract arameters
7	1110	0000	0	0	contract arameters
8	1110	0000	0	0	contract arameters
9	1110	0000	0	0	contract arameters
10	1110	0000	0	0	contract arameters
11	1110	0000	0	0	contract arameters
12	1110	0000	0	0	contract arameters
13	1110	0000	0	0	contract arameters
14	1110	0000	0	0	contract arameters
15	1110	0000	0	0	contract arameters
16	1110	0000	0	0	contract arameters
17	1110	0000	0	0	contract arameters
18	1110	0000	0	0	contract arameters
19	1110	0000	0	0	contract arameters
20	1110	0000	0	0	contract arameters
21	1110	0000	0	0	contract arameters
22	1110	0000	0	0	contract arameters
23	1110	0000	0	0	contract arameters
24	1110	0000	0	0	contract arameters
25	1110	0000	0	0	contract arameters
26	1110	0000	0	0	contract arameters
27	1110	0000	0	0	contract arameters
28	1110	0000	0	0	contract arameters
.....					contract arameters
53	1110	0000	0	0	contract arameters
54	1010	0000	0	0	contract arameters

17.1.3 显示设置 Display setting



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POSITION INDICATOR (SVT Key Sequence M-1-3-5-GOON) 显示设置 Display setting				
At level	PI Display	Left	Right	Remark
0	“-1”	37	1	contract parameters
1	“1”	10	1	contract parameters
2	“2”	10	2	contract parameters
3	“3”	10	3	contract parameters
4	“4”	10	4	contract parameters
5	“5”	10	5	contract parameters
6	“6”	10	6	contract parameters
7	“7”	10	7	contract parameters
8	“8”	10	8	contract parameters
9	“9”	10	9	contract parameters
10	“10”	1	0	contract parameters
11	“11”	1	1	contract parameters
12	“12”	1	2	contract parameters
13	“13”	1	3	contract parameters
14	“14”	1	4	contract parameters
15	“15”	1	5	contract parameters
16	“16”	1	6	contract parameters
17	“17”	1	7	contract parameters
18	“18”	1	8	contract parameters
19	“19”	1	9	contract parameters
20	“20”	2	0	contract parameters
21	“21”	2	1	contract parameters
22	“22”	2	2	contract parameters
23	“23”	2	3	contract parameters
24	“24”	2	4	contract parameters
25	“25”	2	5	contract parameters
26	“26”	2	6	contract parameters
27	“27”	2	7	contract parameters
28	“28”	2	8	contract parameters
29	“29”	2	9	contract parameters
30	“30”	3	0	contract parameters
31	“31”	3	1	contract parameters
32	“A”	10	11	contract parameters
33	“B”	10	12	contract parameters
34	“C”	10	13	contract parameters
35	“D”	10	14	contract parameters

36	“E”	10	15	contract parameters
37	“F”	10	16	contract parameters
38	“G”	10	17	contract parameters
39	“H”	10	18	contract parameters
40	“I”	10	19	contract parameters
41	“J”	10	20	contract parameters
42	“K”	10	21	contract parameters
43	“L”	10	22	contract parameters
44	“M”	10	23	contract parameters
45	“N”	10	24	contract parameters
46	“O”	10	25	contract parameters
47	“P”	10	26	contract parameters
48	“Q”	10	27	contract parameters
49	“R”	10	28	contract parameters
50	“S”	10	29	contract parameters
51	“T”	10	30	contract parameters
52	“U”	10	31	contract parameters
53	“V”	10	32	contract parameters
54	“W”	10	33	contract parameters
55	“X”	10	34	contract parameters
56	“Y”	10	35	contract parameters
57	“Z”	10	36	contract parameters

17.2 Drive parameter setting and monitoring

17.2.1 Drive parameter table

M3-3-1-0					
No.	symbol	default	range	meaning	remark
0	Up level (mm)	000	0~500	Up leveling compensating parameters	
1	Down level (mm)	000	0~500	Down leveling compensating parameters	

M3-3-1-1					
No.	symbol	default	range	meaning	remark
0	Inertia kg/m2	2		system Inertia	
1	Rotate dir 1/0	0	0~1	rotate direction	
2	Pretrq trim (PU)	1.00		gain pre-torque trim	
3	Encoder PPR	02048	0~10000	encoder pulse per revolution	
4	Encoder Sort	1	0~1	encoder Sort	



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5	Encoder dir	0	0	encoder direction	
6	Duty speed mm/s	0400		rated Duty speed	
7	Dely brk lftd ms	0500	0~9999	The brake holding delay	
8	Delay lft brk ms	0500	0~9999	he brake open delay	
9	MTC Sw Pres? 1/0	1	0~1	The motor thermal switch test	
10	Type of motor	0	0~1	Type of main machine	

M3-3-1-2

No.	symbol	default	range	meaning	remark
0	Spdp1 gain	01500	0~10000	gain 1 proportional controller gain1	
1	SpdI1 gain	00500	0~10000	integral controller gain 1 integral controller gain1	
2	ARD Speed (mm/s)	00160	1~500	ARD Speed	
3	ARD Run Direct	00000	0~1	ARD running direction ARD Run Direction	
4	SGP tran21 thr %	030	0~100	SGP convert 21 threshold value SGP transform to 21 threshold	
5	Reset speed %	050	1~100	reset speed Reset speed	
6	feedback filter1	300	0~300	velocity feedback smoothing 1 Speed feedback filter1	
7	feedback filter2	300	0~300	velocity feedback smoothing 2 Speed feedback filter2	
8	Track Error mm/s	0500	0~1000	Speed maximum allowable deviation maximum allowable deviation of speed	
9	ETSC present(%)	00097	1~100	ETSC protection factor ETSC protective coefficient	
10	ETSC Enable(0/1)	00000	0~1	ETSC protection enable ETSC enable protection	
11	overspeed (PU)	1.10	0~1.5	overspeed rate overspeed ratio	
12	ARD Run Mode	0-2	1	emergency rescue mode	



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M3-3-1-3

No.	symbol	default	range	meaning	remark
0	Brk Sw Pres? 1/0	1	0~1	The brake switch test	

M3-3-1-4

No.	symbol	default	range	meaning	remark
0	Rated Power(Kw)	1.2		Rated power of main machine	contract parameters
1	Number of poles	10		Pole number of main machine	contract parameters
2	Rated RPM	153		Rated speed of main machine	contract parameters
3	Rated frq (Hz)	12.8		Rated frequency of main machine	contract parameters
4	Rated voltage(V)	220		Rated voltage of main machine	contract parameters
5	Duty load (kg)	400		Rated load of main machine	contract parameters
6	Rated I (A)	4.6		Rated current of main machine	contract parameters
7	Rated Trq (Nm)	75		Rated torque of main machine	contract parameters

M3-3-1-6

No.	symbol	default	range	meaning	remark
0	Brk dect dlay ms	1000	0~2000	Brake testing time	
1	0mm/s t lim (ms)	0000	0~5000	Zero speed time	
2	Brake settle(ms)	1300	0~5000	The brake releasing time	
3	APR time(ms)	200	100~900	Starting torque compensation time	



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M3-3-1-9

No.	symbol	default	range	meaning	remark
0	Floor number	02	2~55	Floor number	contract parameters
1	Velocity normal	400	0~10000	Elevator rated speed	contract parameters
2	Accelera normal	350	10~1500	acceleration	
3	Jerk0 normal	150	0~1500	Add acceleration 0 Jerk 0	
4	Jerk1 normal	250	0~1500	Add acceleration 1 Jerk 1	
5	Decelera normal	400	10~1500	deceleration deceleration	
6	Jerk2 normal	150	0~1500	Add acceleration 2 Jerk 2	
7	Jerk3 normal	200	0~1500	Add acceleration 3 Jerk 3	
8	Velocity inspect	200	0~640	service speed overhaul speed	
9	Decelera recover	180	10~1500	deceleration Decelerate in nearby floor	
10	Velocity learn	100	0~500	self learning speed learning speed	
11	position gain	20	10~40	position gain	
12	Velocity relevel	30	0~100	Re-leveling speed	
13	Accelera relevel	00300	10~1500	Re-leveling acceleration	
14	Decelera NTSD	01000	10~1500	Deceleration NTSD	
15	Jerk NTSD	00200	0~1500	NTSD Add acceleration Jerk NTSD	
16	pos delay (ms)	00100	0~300	Door area delay	
17	Shv diam (mm)	00400	10~10000	diameter of traction sheave	contract parameters
18	Gear ratio	001.0	1~100	reduction ratio	contract parameters
19	Rope ratio	02	1~7	Rope ratio	contract



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					parameters
20	Run enable	0	0~1	Enable operation	

M3-3-5

No.	symbol	default	range	meaning	remark
0	speed low sect.	00000		The low speed range coefficient of curve segment adjustment	
1	speed mid sect.	00000		The high speed range coefficient of curve segment adjustment	
2	speed Kp1	00010		Proportional controller gain of low speed	
3	speed Ki1	00010		Differential controller gain of low speed	
4	speed Kp2	00010		Proportional controller gain of medium speed	
5	speed Ki2	00010		Differential controller gain of medium speed	
6	speed Kp3	01000		Proportional controller gain of high speed	
7	speed Ki3	01000		Differential controller gain of high speed	
8	speed fliter	00040		Velocity filtering	
9	Pretorque delay	00400		encoder signal starts filtering	
10	Start inner Kp	00000		The proportional controller gain of inner ring	
11	Start inner Ki	00000		The differential controller gain of inner ring	
12	Run source SVT=0	0		command source	
13	Max Drive ErrorB	05		The maximum allowable error time of Class B	
14	Max OVER CURRENT	003		The maximum allowable error time of overcurrent	
15	Start Kp	3500		Start proportional controller gain	
16	Start Ki	0350		Start differential controller gain	
17	VS abnorm t (s)	10.0		V code down time	
18	NTSD Persent %	097		Speed of compel to slow	



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				down	
19	DDP Time (s)	040.0		Time of photoelectric protection in floor	
20	Distance comp	00		Distance compare	
21	Max fdbk error	03		The maximum allowable error time of current feedback	

M3-3-1 (S4-S6-S1-S5)

No.	symbol	default	range	meaning	remark
0	Control methord	3	0~3	Control methord of main machine	
1	MAX DBD ERR	03	0~10	allowable error time of DBD	
2	Rotor pos offset	00000	0~65535	invalid definition	
3	induct d0(mH)	30.00	0.01~99.99	1.8 times of D axis inductance	
4	induct q0(mH)	30.00	0.01~99.99	1.8 times of Q axis inductance	
5	induct d(mH)	30.00	0.01~99.99	D axis inductance,the main machine inductance	
6	mutual resist	30.00	0.01~99.99	Stator resistance	
7	induct q(mH)	30.00	0.01~99.99	Q axis inductance,the main machine inductance	
8	Rotor Time (s)	00.28	0~10	The rotor time	
9	No load current	011.6	0.1~999.9	No load current	
10	Enable time stop	00000	0~1	Restriction of running times	
11	Setuo run times	00000	0.01~99.99	running times setting	
12	Now run times	00000	0.01~99.99	Current running times	
13	Inspect 0stop En	1	0~1	Inspect zero-speed parking functioncan	
14	Reset At PowerOn	0	0~1	Function that the main board resets after losing power	
16	EnableRead Power	1	0-1	The identification function of the base power	
17	INS 6kHz PWM 0/1	1	0-1	Inspect function of down carrier	

M3-3-1 (S7-S4-S1-S2)

No.	symbol	default	range	meaning	remark
0	Drive size	203	0~23	Drive type	
1	Drv Rtd Volt (V)	0220		Rated voltage of drive	
2	Drv I limit (A)	072.0		Hardware limiting current of	



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				drive	
3	AC Line fscale V	1.000	0~2	Input voltage coefficient	
4	Bus fscale (V)	1.000	0~2	Bus voltage coefficient	
5	<3Hz Torque Lmt	2.00	1.5~2.0	Limiting of low current	
6	DC link ovt (V)	425	0~999	Busbar overvoltage value	
7	DC link uv (V)	217	0~999	Busbar undervoltage value	
8	AC Line Input(V)	400	50~480	Rated input voltage	
9	Brake unit work	0.37	0~1	The brake pipe opening voltage	

M3-3-1 (S7-S4-S1-S0)

No.	symbol	default	range	meaning	remark
0	DZ in 2LS	001	0~5	The number of 2LS inside gate area	
1	DZ in 1LS	001	0~5	The number of 1LS inside gate area	
2	Demagnetic (%)	100	0~200	Weak magnetic coefficient	
3	Start filter	01800	0~9999	The encoder signal filter coefficients	
4	Door open speed	100	0~300	Speed of opening ahead	
5	Mtr overload (s)	20	0~40	Time coefficient of motor overload	

M3-3-1 (S3-S6-S9-S8)

No.	symbol	default	range	meaning	remark
0	H_speed frq (Hz)	10000		High-speed switch frequency	
1	Min IGBT on t us	000.0	0~5.0	The minimum allowable narrow pulse	
2	Pos Err Lim (mm)	0200		Trigger distance of position lost	
3	Switch frq (KHz)	10		frequency switch frequency	
4	Run time hour	00000		Run time hour	
5	Run time minutes	00		Run time minutes	

17.2.2 驱动监控菜单 Drive monitoring menu

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No.	symbol	range	meaning	remark
0	Software version		Software version	
1	Car speed mm/s		Car speed	
2	Motor Speed RPM		Motor Speed(Based on PG)	
3	Rotor position		Rotor position(machinery corner)	
4	Dictated v mm/s		Given speed(DBSS)	
5	Output voltage V		output voltage	
6	Mtr trq PU		given torque current (relative torque current)	
7	Output current A		current Output current	
8	Heat sink temp1		Temperature of heat sink 1	
9	Heat sink temp2		Temperature of heat sink 2	

M3-1-2

No.	symbol	range	meaning	remark
0	DC link V		Dc busbar voltage	
1	X offset (A)		X phase current sensor offset	
2	Y offset (A)		Y phase current sensor offset	
3	Z offset (A)		Z phase current sensor offset	
4	Frequency (Hz)		Output frequency	
5	speed gap (mm/s)		Speed gap	
6	LS distance (mm)		distance that compulsive deceleration should be installed (corresponding to the parameters)	
7	Motor overload		Integrator of motor overload	
8	Drive overload		Integrator of inverter overload	

M3-1-3

No.	symbol	range	meaning	remark
0	Encoder pulse		The encoder pulse condition	
1	Encoder Z num		The encoder Z phase count	
2	Encoder C POS		The encoder C phase position	
3	Encoder D POS		The encoder D phase position	
4	Factory resd1		Factory test 1	
5	Factory resd2		Factory test 2	
6	Factory resd3		Factory test 3	
7	Factory resd4		Factory test 4	



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8	Factory resd5		Factory test 5	
9	Now floor		current floor	
10	Now position (m)		current position	
11	INPUT1		INPUT1	
12	INPUT2		INPUT2	
No.	symbol	range	meaning	remark
13	OUTPUT		OUT	
14	OUTPUT CODE		OUT CODE	
15	Prepare to run		Prepare to run(by hand or DBSS from MCSS)	
16	LOAD Percent %		The actual weight compensation	
17	Bases Enabled		DBSS status	

M3-1-4

No.	symbol	range	meaning	remark
0	2LS (m)		Position of up compulsive deceleration 1	
1	4LS (m)		Position of up compulsive deceleration 2	
2	1LS (m)		Position of down compulsive deceleration 1	
3	3LS (m)		Position of down compulsive deceleration 2	
4	Floor 1A (m)		Position of 1 st floor after hiostway learning	
5	Floor 1B (m)		Position of 1 st floor after hiostway learning	
6	Floor 2A (m)		Position of 2 nd floor after hiostway learning	
7	Floor 2B (m)		Position of 2 nd floor after hiostway learning	
...	...		...	
...	...		...	
110	Floor 54A (m)		Position of 54 th floor after hiostway learning	
111	Floor 54B (m)		Position of 54 th floor after hiostway learning	
112	Floor 55A (m)		Position of 55 th floor after hiostway learning	



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113	Floor 55B (m)		井 Position of 55 th floor after hiostway learning	
114	NOW PULSE(1000p)			

18. Common failures and troubleshooting methods

18.1 If mainboard breaks code,then display the current fault code

The warning code	Operating mode	Fault code	Operation mode or the drive failure
AL000	EPC(Emergency power control mode)	ER100	DTC(The door in the setting time can't be locked completely)
AL001	COR(Reset mode)	ER101	DTO(The door in the setting time can't open completely)
AL002	EFS(Emergency Firefighters service mode)	ER102	NAV(The elevator is not available
AL003	EFO(Emergency Fire operation mode)	ER103	DBP-Fault(The door branch fault)
AL004	EQO (Emergency earthquake operating mode)	ER104	TCI-Lock(The car roof repair activation is locked)
AL005	EPR(Emergency rescue operation mode)	ER105	LS-Fault(Compulsive deceleration lost)
AL006	EPW(emergency pattern Waiting for rescue)	ER106	1LS+2LSon (up and down compulsive deceleration act at the same time)
AL007	OLD(Overload)	ER107	Adr.- Check(address detected)
AL008	ISC(independent Service)	ER108D BSS-Fault	DBSS-Fault(Drive not ready)
AL009	ATT(Attendant Service)	ER109	SE-Fault (The thermal switch of main machine fault)
AL010	CTL(Cars to specific floor model)	ER110	start DCS!(Door test sequence is not running)
AL011	CHC(Shut down the hall call mode)	ER301	IGBT Fault



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AL012	LNS(Load non stop)	ER302	OVER Current (actuator over current fault)
AL013	MIT(Moderate passenger traffic model)	ER304	Overtemp (heat sink is over temperature)
AL014	DCP(cars protection delay mode)	ER305	Motor overload
AL015	ANS(Anti-nuisance car call protection)	ER306	Drive overload
AL016	ARD(Automatic Rescue Device)	ER307	DC busbar overvoltage
AL017	PKS(Parking Service)	ER308	DC bus undervoltage
AL018	GCB(Button control mode)	ER309	Motor overspeed
AL019	EHS(Emergency medical service mode)	ER310	PVT lost (Encoder lost)
AL020	ROT(Disturbance model)	ER312	Base power read failure
AL021	INI(System initialization mode)	ER313	Task overrun(Pwm interrupt task execution timeouts)
AL022	INS(inspection mode)	ER314	Tune Moving(Encoder angle self-learns displacement of main machine)
AL023	ESB(Safety circuits failure mode)	ER315	Track error (Speed follow the failure)
AL024	DHB(Door hold mode)	ER316	1LV NORMAL CLOSE (1LV Photoelectric fault)
AL025	ACP(crime prevention)	ER317	2LV NORMAL CLOSE (2LV Photoelectric fault)
AL026	WCO(Cars savage operate, hall calls are invalid)	ER318	Floor number(Floor number error)
AL027	DBF (DBSS fault)	ER319	PARAMETER ABNORMAL(parameter Setting of main machine faults)
AL028	SHO(The Sabbath mode)	ER325	E2 write err (Eeprom, reading and writing error)
AL029	DAR(failure mode of DBSS, which automatic return to the lowest stopping floor, hydraulic)	ER326	Motor thermal protection (motor thermal relay action)
AL030	DCS(Inspect door sequential mode)	ER328	Brake dropped (Brake switch 1 test failure)
AL031	UFS(The upper limit switch triggers DS3 mode)	ER329	AC Line imbalance (Three phase output current is unbalanced)



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AL032	ACC(Accelerate to NOM SPE or SHR SPE mode)	ER330	AC Line UVT (Three phase output power is undervoltage)
AL033	DoorBridge,	ER331	Brake Unit Fault
		ER332	Thyristor module (Thyristor feedback fault)
		ER333	DBD PICKUP(main contactor or brake contactor action failure)
		ER334	DDP Fault (photoelectricity is not detected in setting time)
		ER335	Brake dropped2 (Brake switch 2 test failure)
		ER342	Inv Ix Offst (U phase current sampling failure))
		ER343	Inv Iy Offst (V phase current sampling failure)
		ER344	Inv Iz Offst (W phase current sampling failure)
		ER347	Base AD Offst (0.5 V / 1.5 V reference voltage fault)
		ER348	ETSC relay fault
		ER361	Vcode 1 is abnormal
		ER362	Vcode 2 is abnormal
		ER363	Power Lost

18.2 The common logic fault table

Enter into the M1-2 for logic fault, and turn to the current logical failure. Press 1 + 2 can eliminate the current failure. Logic fault can be continuously deleted if continuously press 1 then press 2.

The event	The possible reasons	Associated Settings
total runs	The total times of running from last time on power	-
minutes on	The total minutes of running from last time on power	-
The flashing message		
The event	The possible reasons	Associated Settings
LS - Fault	signal of up compulsive deceleration 2 LS and down compulsive deceleration 1 LS is abnormal	
1LS + 2LS on	Compulsive deceleration switch 1LS and 2LS act at the same time	-



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DBSS - Fault	Inverter not ready yet	-
TCI - Lock	Operating sequence of car roof inspection switch is not right, must open the hall door in the following order. 2. The car roof inspection switch should turn inspection to normal condition. 3. Close the hall door If not operating in the above order, the elevator will not be able to run normally, at the same time, INS indicator light on logic control board will be flashing.	-
DBP - Fault	The elevator operation must be done in the correct order If door area signal is not inspected when the elevator stops, the possible reason is breakdown of LVC relay This error will be stored on the logic control part of the EEPROM, only can use INS maintenance switch to operate on the motor.	
Adr - Check	Address of special remote station is not correspond to the standard I/o list	TCI (691) = 16 - 3 1LS (692) = 16 - 1 2LS (693) = 16 - 2
start DCS!	a) Normal operation is not allowed if operation of door checking start (DCS) is not complete b) On condition of normal operation and that the elevator stops at door area with car door fully opening (DOL is valid), full close signal of the hall doors is triggered (DW is valid). Then carefully check whether hoistway hall door lock is short-circuited. Start DCS according to the M-1-3-6 ! Note: if you want to cancel the DCS and directly into the M1-9, the address should be changed to 200095 c0: FF (please use this operation prudently)	NoDW - Chk
OCSS		
The event	The possible reasons	Associated Settings
0100 OpMode NAV	The breakdown of Drive part will cause inoperable of OCSS. This model will also be triggered after maintenance or before inspection	-



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0101 EPO shutd.	Cars can't run in EPO mode.	NU (017) NUSD (018) NUSG (019)
0102 OpMode DTC	Door can not normally close within the setting time (DCL, DFC or DW lost).	DCL (694) RDCL (695) DOOR, REAR DTC-T
0103 OpMode DTO	Door can not normally open within the setting time	DOL (000) RDOL (544) DOOR, REAR DTO-T
0104 OpMode DCP	Car can't respond calls or instructions in setting time (such as the door is blocked by an obstacle).	DCP-T
0105 DBSS fault	Drive failure	DRIVE
0106 PDS active	Some door switches are not closed.	PDD (784)
0110 BrakeSlfErr	Brake braking force monitoring ineffective	
0112 UCM FAULT	Car unintentional movement production	DRIVE=1
MCSS		
The event	The possible reasons	Associated Settings
0200 Pos. Count.	After complete operation, the system finds count of door area and IP signal is not match, may be sending time of door area signal DZ and IP signal is too short, system doesn't have time to inspect.	LV-MOD, DZ-TYP
0201 correct. run	Adjusting operation (such as electricity, maintenance, after NAV. etc.)	-
0202 /ES in FR	ES signal is activated.when the elevator is at high speed	MD/AES, ES-TYP
0203 /ES in SR	ES signal is activated.when the elevator is at low speed	MD/AES, ES-TYP
0204 TCI/ERO on	TCI or ERO switch is been acted	ERO-TYP
0205 SE-Fault	The elevator can't start due to loss of SE signal (check SKL, considered, fuse, etc.)	-
0207 DDP in FR	The shaft signal can't be detected within the setting time (DDP) when elevator is at high speed (DZ signal lost).	DDP
0208 DDP in SR	The shaft signal can't be detected within the setting time (3P) when elevator is at low speed (DZ signal lost).	3P

0209 DDP in RS	The shaft signal can't be detected within the setting time (3P) when elevator starts rescue operation (DZ signal lost)..	3P
0210 /DZ in NST	The shaft signal can't be detected when the elevator stops	LV-MOD, DZ-TYP
The event	The possible reasons	Associated Settings
0211 /DFC in FR	The safety circuit breaks when elevator is at high speed	-
0212 /DFC in SR	The safety circuit breaks when elevator is at low speed	-
0216 DZ missed	The UIS and DIS signal has been detected but DZ is not received yet, the possible reason is failure of LV relay.	EN-RLV, DRIVE
0224 J-Relay	Logic control part detected that three-phase power supply is fault (such as lack of phase or wrong phase)	EN-J, J-T
0226 LS-fault	Signal of compulsive deceleration is not normal, see flashing information.	-
0228 1LS+2LS on	The signal of 1LS and 2LS is detected at the same time	1LS (692) 2LS (693) DRIVE C-TYPE
0230 RSL Adr chk	See flashing information description	NoAdrChk
0231 LSVF-W:/DR	Frequency converter fault (not ready)	DRIVE C-TYPE (see MCB)
0232 LSVF-W:/SC	If elevator is too fast during deceleration, it can't acts function of opening the door in advance, ADO function, or RLV >.	(see MCB)
0237 /DW in FR	The hall door loop disconnect when elevator runs fast.	-
0238 /DW in SR	The hall door loop disconnect when elevator runs slowly.	-
DCSS		
The event	The possible reasons	Associated Settings
0300 DBP: dfc_SE	When the door is opening or door has fully opened DFC or SE (with ADO) not acts.	EN-RLV DRIVE

0301 DCL in []	DCL signal is detected when the door is fully open	DCL (694) RDCL (695)
0302 DCS:DW err	During normal operation, hall door back closure when the door is open.	NoDW-Chk
0303 DBP-Fault	See flashing signals' DBP - Fault '	-
0304 DOL:alw. on	DCL signal is detected when the door is fully closed; If the fuse of DO2000 door machine breaks will also appear this malfunction.	DOL (000) RDOL (544)
SSS		
The event	The possible reasons	Associated Settings
0400 RSL parity	Two remote stations connected to the same serial line are using the same address.	-
0401 RSL sync	Sync signal of remote serial line lost.	-
GROUP		
The event	The possible reasons	Associated Settings
0500 RNG1 msg	Data in the serial loop of parallel/group control is wrong.	-
0501 RNG1 time	In a certain period of time, does not receive signals from the other lifts.	GROUP
0502 RNG1 sio	Serial circuit transformat of parallel /group control fault.	-
0503 RNG1 tx	Serial data transmission is overtime.	-

18.3 Common drive malfunction table

Enter into M3-2-1 for the current drive failure, into M3-2-2 for historical drive failure, into M3-2-3 to eliminate current drive failure, into M3-2-4 to eliminate all drive failure, into M3-2-5 for running time on power

M3-2-5 running time on power - M3-2-2 historical failure time = Failure occurs from the current time

No.	Fault display	Fault	Class	Fault system and the trigger condition	The possible reasons
1	IGBT Fault	Module fault	B		A, the drive output side U, V, W is short circuit;

				It will run in condition that module is over current, output is short circuit, or the brake does not open. The trigger signal is from IPM module IGBT module fault	B, the brake is not open, while the drive has output current, C, the locks or safety circuit is suddenly broken when the elevator is in the normal operation, maintenance, self-learning, and correction D, IGBT module is damaged E, drive and mainboard line in poor contact
2	Over current	Actuator over current fault	B	The operation will be triggered in condition that module is over current, output is short circuit, or the brake does not open, or In condition that the load is too heavy, current is too large, speed up or slow down too quickly, the drive specification is too small, module is damaged and damage is damaged. Trigger signal is from the current detection path, hall sensor. Over current	A, the drive output side U, V, W is short circuit; B, the brake is not open, while the drive has output current, C, the locks or safety circuit is suddenly broken when the elevator is in the normal operation, maintenance, self-learning, and correction D, IGBT module is damaged E, cable is damaged; F, orientation angle is not correct; G, motor runs without phase J, encoder signal lost
3	Motor overload	Motor overload	A	When the control panel CPU through the integration of the output current, integral accumulator exceeds a certain value will trigger the motor overload fault.	A, The brake doesn't open when the elevator runs B, orientation angle of main machine is not correct C, overload doesn't work, the elevator is overload in fact D, The setting of protective coefficient is too small
4	Drive overload	Drive overload	A	When the control panel CPU through the integration of the output current, integral accumulator exceeds a	A, The brake doesn't open when the elevator runs B, orientation angle of main machine is not correct C, overload doesn't work, the

				certain value will trigger the rive overload fault.	elevator is overload in fact D, The setting of protective coefficient is too small
5	Overtemp	Radiator is overtemp	A	Module is overtemp, either of two temperature testing point, including IPM and rectifier bridge, temperature is >85 °C	A, temperature sensor is damaged; B, cooling fan of drive is damaged; C, the actual heat sink achieves this temperature;
6	DC link OVT	DC busbar is overvoltage	A	1, when the control panel CPU detects signals from hardware is overvoltage will trigger overvoltage fault. 2, when the control board CPU detects that busbar voltage is higher than the setting value of software point will trigger overvoltage fault. DC bus voltage is overvoltage	A, braking resistor disconnect; B, external input voltage is too high; C, control panel with electricity first, then the drive on electricity, the impact may cause overvoltage fault
7	DC link UVT	DC busbar is undervoltage	A	1, when the control panel CPU detects signals from hardware is undervoltage will trigger undervoltage fault.. 2, when the control board CPU detects that busbar voltage is more than the undervoltage point which software set will trigger undervoltage fault DC bus voltage is undervoltage	A, input LN power supply voltage is under 207 v; B, base without electricity; C, the setting of parameter "Bus fscale (V)" is wrong; D, rectifier bridge, or soft start resistance fault;
8	Overspeed	main machine	B	The speed of main	A The setting of running speed

		is overspeed		machine is faster than the rated speed (M3312) times over speed	"Velocity normal" is too big; B, The setting of the inertia "Inertia kg/m2" is too big; C, The setting of encoder related parameters, tractive sheave diameter, reduction ratio, rope ratio is not correct; D, no output from frequency converter;
9	Track error	Speed track error	B	If the actual motor speed which control board CPU detected and the speed of setting satisfy the following two conditions at the same time, will trigger speed track error. 1) $\text{abs}(\text{actual elevator speed} - \text{given speed}) > \text{rated speed} \times \text{Track error} / 100$ (2) the duration is longer than the setting time value	A, The brake doesn't open when the elevator runs B, orientation angle of main machine is not correct C, main machine runs, encoder does not move; D, the encoder signal is not correct E, The setting value of track error is too small, may occurs when maintenance operation
10	PVT lost	Encoder fault	B	The encoder signal lost, abnormal or encoder hardware failure (such as power problem, line breaks etc.)	A, the encoder line doesn't connect or breaks B, the motor phase sequence is not correct or the direction setting of encoder is not correct when positioning C, the setting of encoder pulse count and type is not correct D, encoder line and shielding line are short circuit
11	Task orun	The operation	B	The operation of interrupting 1 ms, 10	A, The setting of PWM carrier

		of interrupting Pwm is overtime		ms and 40 ms is overtime, an interrupt cycle is not fully perform all the above tasks	frequency "Switch FRQ (KHz)" is too big;
1 2	Tune Moving	Encoder angle self-learns displacement of main machine	B	During self-learning of pole and encoder angle, the rotor moves, then cause the failure of self learning	A, The main machine is not fully clasped because the brake clearance is too large B, encoder is not fixed;
1 3	E2 write err	read-write of Eeprom is wrong	B	The control board CPU will operate read-write of EEPROM when self-inspects with electricity. We can judge whether EEPROM is normal operates according to the read-write of special unit.If the data of read-write is inconsistent for three times, fault will be triggered. In the normal operation with electricity, if there is an EEPROM writing operation, write in first, and then read again.It is normal if data of reading and writing is the same, otherwise,consistently operates for 3 times, if still not the same, would trigger fault.	A, EEPROM chip is damaged
1 4	Mtr therml cntct	The action of motor thermal relay	D	The protection will start when control board CPU detects the	A, the action of motor thermal relay (temperature of main machine is too high, thermal relay is

				signals of motor thermal relay action.	damaged); B, connection error (such as: line doesn't connect)
1 5	Brake dropped	State failure of brake switch 1	B	Control board CPU continuously inspect the status of the brake switch, and compare the command from the software for status of brake with the actual status from detecting. During the test time of brake switch, the continuous inconformity of these two statuses would trigger fault. Includes the following two situations: (1) can't trun on in runtime; (2) can't shut down when machine stops	A, the actual action of brake switch doesn't conform to the setting of fault B, line connection is wrong; C, the setting of brake switch testing time is too small (default 500 ms);
1 6	Brake dropped 2	State failure of brake switch 2	B		
1 7	Brake Unit Fault	The braking unit fault	A	When the inverter is braking, in the case of no other faults, if detect that FIGT signal is low electric frequency, then trigger the fault.	A, brake IGBT is damaged B, braking resistor short-circuit;
1 8	Thyristor module fualt	Thyristor fault	A	Drive plate has the hardware circuit to monitoring state of thyristor. In normal situation, the voltage drop of thyristor's anode and cathode is very small, when thyristor can't work, the voltage drop will be very big. In this case	A, thyristor is damaged

				for a long time, the buffer resistance will be damaged. The hardware circuit monitors the pressure drop for fault detection.	
19	DBD PICKUP	Main contactor or brake contactor fault	B1	Control board CPU continuously inspects the input state of DBD, and compares the command status from the software for contactor control and brake control with the actual input status of DBD from detecting. If the inconsistent duration of these two statuses is more than 100 ms, would trigger fault. Decision for consistent status is as follows: 1) either of UDX and LB outputs, DBD inputs 0; (2) neither of UDX and LB outputs, DBD inputs 1; Set maximum failure times in MAX DBD ERROR..	A, the actual status of LB or UDX is inconformable with the setting of fault B, power supply for LMCB board (DC24V) first, then DC30V C, short circuit UDX and LB when main machine locates, however, the RUN of the M - 3-3-5 SOURCE is set to 1 (automatic);
20	DDP Fault	Photoelectric disconnect	A	Control panel when the CPU will run in normal operation and reset detection must through signal, when in a set of time without must pulse signal, will	A, no signal is inputted as photoelectric switch is damaged, B, The setting of DDP time is too small, so this fault may occur in the reset operation (half of rated speed). (In this case, move compulsive deceleration switch to two sides, or shield DDP, and set

				trigger fault.	M335 DDP time to 0.)
2 1	Vcode abnormal1	Vcode abnormal1	C	The elevator stops, and not responds for the V code from logic for more than 100 * "VS abnorm t (s)" ms	A, During the debugging,if the elevator is not on the leveling, and no maintenance, then the setting of RUN enable is 0; B, Run enable is not set to 1 after the completion of hoistway self-learning C, at high speed running, DFC signal regains after suddenly lost
2 2	Vcode abnormal2	Vcode abnormal 2	C	Drive in zero speed is not on leveling, and logical CPU continuously sends the SD code of stopping on the next leveling as it's not on leveling now.If the control board CPU continuously detects this state for morethan 6 seconds, will trigger fault.	A, when running at low speed, break the door lock; B, no safety circuit boards, and set function of opening in advance;
2 3	Power lost	Main power lost	C	Busbar voltage will decreases after power lost. Fault will be triggered when voltage is below DC380V, and fault disappears automatically when power rises and voltage of busbar is higher than DC400V.	A, power off/on normally B, control panel is on power, while drive is not C, platoon line is not connect
2 4	1LV NORMAL CLOSE	1LV photoelectricity fault	A	1 LV photoelectricity signal lasts till elevator runs 1000 mm	
2 5	2LV normal closed	2LV photoelectricity fault or the position of elevator lost	A	2LV photoelectricity signal lasts till elevator runs 1000 mm, or floor number is not the same with detected	The photoelectricity has no change after running 200 mm from the leveling position

				photoelectricity number during running, or the difference between photoelectric position of elevator in and out and position of self learning is larger than 1000 mm	
2 6	Floor number err	Floor failure	A	Elevator self learning is complete, and on condition of 2LS, the floor from learning is not the same as the setting floor height	Setting of floor parameter error
2 7	PARA. ABNORMA L	Parameter setting fault	A	The setting frequency of motor, rotate speed of motor and number of pole-pairs does not match the formula $N = 60 f/P$	
2 8	Current AD Offst	When machine stops, current of any phase is too large	B	When machine stops, current of UVW phase is too large	A, fault of current sampling channel, hardware fault of mainboard or driver board; B, current Hall sensor fault; C, eight of UVW phase shorts circuit to ground

18.4 Related fault of integrated ARD (automatic emergency device)

No.	The fault phenomenon	The reason and investigation of the problem
1	Power off, UPS starts, but the system does not enter into the EPR model	1, parameter setting problem, IO 17 value of ALMCB board (NU: Emergency power) should be set to logic = 0, address = 1, bit = 0 2, the hardware circuit problem, at this moment, the voltage of motherboard P1-3 should be DC24V.If no voltage, will report

		the power lost fault, please check the circuit of PW contactor and power supply circuit 3, Make sure all air switches in control cabinet are in the access status, otherwise can't enter into ARD
2	Power off, UPS doesn't work	1, UPS starts while switch doesn't 2, UPS battery is not enough, or is damaged 3, check input terminal of UPS whether is AC220V. If the input power serves, UPS will turn into mode of accumulator outputting
3	Power off, ARD can't work	1, UPSC contactor is damaged, coil does not work 2, UPSC contactor close, check whether UPSC normally open contact 3, 4, 5, and 6 is closed, otherwise drive doesn't have power supply
4	Power off, mainboard does not work	1, UPSC contactor is damaged, coil does not work 2, UPSC contactor close, check whether UPSC normally open contact 53, 54, 63, and 64 is closed, otherwise switching power supply without power, resulting in mainboard no power
5	rescue direction of ARD is inconsistent with the actual direction of car load	1, If car side heavy then ARD's direction is downward, otherwise on the contrary. If the direction is wrong, correct parameters ARD Run Direct = 0/1 2, ensure the encoder connection is all right, phase sequence of UVW is correct, this can meet the normal condition of fast car
6	ARD rescue leveling is bad	1, setting of ARD speed is too large, the parameters scope should be 1-500, we advise to set the range to 80-120 2, Spacing of two photoelectricities is not installed in place, the propositional spacing is 60 mm 3, photoelectric signal is abnormal, please check photoelectric signal connection

7	Scram during ARD rescue process	1, safety loop disconnects or lock loop disconnects, should recover circuit 2, control cabinet or car repairs, when recover maintenance to normal, elevator continue ARD 3, power of UPS is not enough, ensure accumulator is charged enough when you use for the first time. 4, Type of UPS is not correct. Select UPS of capacity 1KVA/800W and type C1K 1KVA/800W for main machine power $\leq 11.7\text{kw}$, the brake power $\leq 250\text{W}$ and no brake forced function; Select UPS of capacity 2KVA/1600W and type C2K 2KVA/1600W for main machine power 11.7kw-25kw and the brake power $\leq 500\text{W}$ 5, Report the failure of drive. Server monitors fault menu, and eliminates according to the type of fault
8	the door does not open when elevator moves to leveling with ARD	1, the DDO dial-up in mainboard is “off”, it should be “on” 2, signal wire connection of doors open and close is incorrect, please check whether wiring harness of doors open and close is correct according to the principle diagram 3, signals of virtual layer are detected, please check setting of virtual layer signal 4, Parameter setting of EPO - DC is too small, this parameter is for closing time when door opens fully during rescue. Please set this parameter according to actual requirement.
9	the door does not open fully when elevator moves to leveling with ARD	1, signal that the door opens fully is not correct, please check the wiring and parameter setting of related IO 2, Wrong signal of the back door is detected, and back door opens. Please check signal location of magnetic stripe in back door. 3, If door is jammed, check the door machinery.
10	ARD starts again after scrambling	1, control cabinet or car repairs, when recover maintenance to normal, elevator continue ARD 2, If power off immediately after on power, ARD continues

11	When on power, ARD can't convert to normal mode	1, PW contactor is damaged, so that drive is not on power. 2, check whether normal close point of the PW contactor 71, 72 is disconnected, it should be disconnected with normal power supply
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18.5 Other system failure

Running state failure

No.	Fault description	The possible reasons	Method of Confirmation
1	The elevator scrams when running fast. It can't continue to run even if not on the leveling	1, The safety signal interrupts in running; 2, The door lock signal interrupts in running 3, The system faults in running	1. If ES of main board lights on, check M112 by server. ES signal capitalized means the safety circuit disconnects 2. If DFC or DW of main board lights off, check M112 by server., DFC and DW signals lowercase 3. If ERR trouble light always on, check current fault code of the M321. If ERR trouble light is not bright, check the fault records of M322 or M121 logic fault
2	The elevator scrams In operation of maintenance, it can't continue to run	1, The safety signal interrupts in running; 2, The door lock signal interrupts in running 3, The system faults in running	1. If ES of main board lights on, check M112 by server. ES signal capitalized means the safety circuit disconnects 2. If DFC or DW of main board lights off, check M112 by server., DFC and DW signals lowercase 3. If ERR trouble light always on, check current fault code of the M321. If ERR trouble light is not bright, check the fault records of M322 or M121 logic fault

3	The elevator scrams In operation of hoistway self-learning	1, The safety signal interrupts in running; 2, The door lock signal interrupts in running 3, The system faults in running 4, The setting of number of floors is wrong	1. If ES of main board lights on, check M112 by server. ES signal capitalized means the safety circuit disconnects 2. If DFC or DW of main board lights off, check M112 by server. DFC and DW signals lowercase 3. If ERR trouble light always on, check current fault code of the M321. If ERR trouble light is not bright, check the fault records of M322 or M121 logic fault 4. Check the TOP value of M1311 and the Floor number value of M3319
4	The elevator scrams in the operation of restoration. It can't continue to run even if not on the leveling	1, The safety signal interrupts in running; 2, The door lock signal interrupts in running 3, The system faults in running 44, photoelectric switch faults in the operation that it outputs all the time. Elevator stops when meets signal of compulsive deceleration (1/2LS); 5, position signal (1/2LS) faults in the operation that it outputs all the time. Elevator stops when meets photoelectric signal	1. If ES of main board lights on, check M112 by server. ES signal capitalized means the safety circuit disconnects, check the action of each safety switch 2. If DFC or DW of main board lights off, check M112 by server., DFC signal lowercases means that the hall door lock disconnects while DW signal lowercases means that the car door lock disconnects 3. If ERR trouble light always on, check current fault code of the M321. If ERR trouble light is not bright, check the fault records of M322 or M121 logic fault 4. If 1/2 LV signal of M112 is capital no matter on leveling or not, then photoelectric switch faults 5. If 1/2 LV signal of M112 is capital no matter in position of compulsive deceleration or not, then 1/2LS is abnormal

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5	The elevator goes above overhead	1, Signal of compulsive deceleration (1/2LS) faults 2, photoelectric signal faults that no output; 3, Slip distance is too large when starts; 4, Slip distance is too large when stops; 5, the position of UIS/DIS is opposite,so the elevator goes to overhead when releveing	1. Run the elevator at low speed to make sure that signal of compulsive deceleration M112 (1/2LS) is capital or lowercase all the time before the elevator goes to the actual position of compulsive deceleration, then signal of compulsive deceleration(1/2LS) faults 2. When elevator arrives to the bottom/top leveling, 1/2LV signal of M112 does not turn into capital 3. Check whether slip distance will cause superior limit switch to act when the elevator starts on overhead 4. Check whether slip distance will cause superior limit switch to act when the elevator stops on ground floor 5. When elevator runs slowly, check whether the action sequence of UIS / DIS signal in the M112 is correct.	
6	电梯蹲底 The elevator goes to pit	1, Signal of compulsive deceleration faults; 2, photoelectric signal faults that no output; 3, Slip distance is too large when starts; 4, Slip distance is too large when stops; 5, the position of UIS/DIS is opposite, so the elevator goes to pit when releveing	1. Run the elevator at low speed to make sure that signal of compulsive deceleration M112 (1/2LS) is capital before the elevator goes to the actual position of compulsive deceleration. If lowercase all the time, then signal of compulsive deceleration(1/2LS) faults 2. When elevator arrives to the bottom/top leveling, check whether 1/2LV signal of M112 is capital 3. Check whether slip distance will cause high-low limit switch to act when the elevator starts on overhead 4. Check whether slip distance will cause high-low limit switch to act when the elevator stops on ground floor 5. When elevator runs slowly, check whether the action sequence of UIS / DIS signal in the M112 is correct.	
7	In operation of maintenance, main machine sounds strange,	Drive faults;	Check current fault code of M321. If no, then check the fault records of M322	

	ERR lights		
8	Speed activated when the elevator maintains, and can't run again	1, drive faults; 2, encoder line is disconnected;	1. Check the current fault code of M321 (2) Check whether lines on both ends of encoder is connected by the multimeter
9	Press the maintenance button of up/downward, the elevator doesn't move	1, no safety signal; 2, no door lock signal; 3, no signal of closed in place; 4, setting of function parameter is incorrect 5, system faults;	1, Check M112 by server. ES signal capitalized means the safety circuit disconnects, check the action of each safety switch 2. Check M112 by server. If DFC signal lowercases means that the hall door lock disconnects while DW signal lowercases means that the car door lock disconnects 3. On the condition that both DFC and DW of M112 are capitalized, signal of DCL closed in place is lowercase. 4. RUN source SVT of M335 should be 0 5. If ERR trouble light always on, check current fault code of the M321. If ERR trouble light is not bright, check the fault records of M322.
10	Press the upward maintenance button, the elevator goes up while press the downward button, the elevator doesn't move	1, the control panel doesn't receive the order of moving down; 2, There is 1LS/photoelectric signal at the same time;	1. When press the downward maintenance button, DIB of M112 is not capital 2. Check whether 1LS and 1LV of M112 is capital
11	Press the downward maintenance button, the elevator goes	1, the control panel doesn't receive the order of moving up 2, There is 2LS/photoelectric signal at the same time	1. When press the upward maintenance button, UIB of M112 is not capital 2. Check whether 2LS and 2LV of M112 is capital

	down while press the upward button, the elevator doesn't move		
12	When press the downward maintenance button, the elevator goes down while press the upward button, the elevator stops after moving some distance	1, drive faults; 2, , DC busbar is over-voltage due to fault of braking resistor	1. Check current fault code of M321. If no, then check the fault records of M322 2. Remove the brake resistance wire, and measure resistance value by multimeter. If the resistance is damaged, there is very large deviation between the measured resistance value and standard resistance on braking resistance box
13	When press the upward maintenance button, the elevator goes up while press the downward button, the elevator stops after moving some distance	1, drive faults; 2, , DC busbar is over-voltage due to fault of braking resistor	1. Check current fault code of M321. If no, then check the fault records of M322 2. Remove the brake resistance wire, and measure resistance value by multimeter. If the resistance is damaged, there is very large deviation between the measured resistance value and standard resistance on braking resistance box
14	The elevator only can maintain,can't self-learning	1, safety loop disconnect on normal condition; 2, Run enable is not set to 0; 3, the system is in maintenance mode;	1. The ES signal of M112 is capital, while DFC or DW is lowercase 2 Ensure that Run enable of M3319= 1 3. Status display of M111 is INS, and ERO of M112 is capital
15	After the self learning, the elevator can't be called by server tool or hall call	1, Run enable is not set to 0; 2, not start the DCS, or not cancel it; 3, If EN - CRT = 0, it can't run above 16th floor; 4, system is not in the state of IDLE;	1. Ensure that Run enable of M3319= 0 2. The current status display of M111 is Start DCS 3. When the elevator is higher than 16 th floor, EN – CRT of M1310 = 0 4. elevator state of M111 is not IDLE

16	Press the maintenance button, the elevator always slips to the heavy side	1, improper torque compensation; 2, track err function is blocked;	1. Confirm value of KP and KI in M335, and adjust it 2. Track error of M3312 is set to 0
17	In normal operation, the elevator slips 2~4cm to the heavy side, then to the specified direction.	1, improper torque compensation; 2, the J11 / J12 jumper of mainboard is connected to the right or the J11 / J12 is not installed;	1. Confirm value of KP and KI in M335, and adjust it 2. the J11 / J12 jumper of mainboard is connected to the right or the J11 / J12 is not installed
18	The elevator revises operation after arriving to the target floor	System faults;	Check the fault records of M322 or M121 logic fault
19	The elevator suddenly stops during normal running at high speed, then revises operation	1, The safety signal interrupts in running; 2, The door lock signal interrupts in running 3, The system faults in running	1, If ES of main board lights on, check M112 by server. ES signal capitalized means the safety circuit disconnects, check the action of each safety switch 2. If DFC or DW of main board lights off, check M112 by server., DFC signal lowercases means that the hall door lock disconnects while DW signal lowercases means that the car door lock disconnects 3. check the fault records of M322 or logic fault of M121
20	The elevator goes above overhead when up reset	1. switch of up compulsive deceleration doesn't work or installation distance is too short 2. The board of overhead in the wrong place 3. The photoelectricity faults	1. 2LS of M112 lowercases all the time and doesn't work or the distance between action position and the overhead leveling is too short (distance of 1 m/s should be 850 mm, distance of 1.5 m/s should be 1750 mm, distance of 1.75 m/s should be 2200 mm) 2. The board of overhead is too close to the up limit position (standard is 150 mm) 3. When leveling changes, 1 lv and 2 lv of M112 does not become into capital

21	The elevator goes to pit when down reset	1. switch of down compulsive deceleration doesn't work or installation distance is too short 2. The board of ground floor in the wrong place 3. The photoelectricity faults	1. 1. 2LS of M112 lowercases all the time and doesn't work or the distance between action position and the bottom leveling is too short (distance of 1 m/s should be 850 mm, distance of 1.5 m/s should be 1750 mm, distance of 1.75 m/s should be 2200 mm) 2. The board of ground floor is too close to the up limit position (standard is 150 mm) 3. When leveling changes, 1 lv and 2 lv of M112 does not become into capital
22	The elevator runs to the target floor, but it is higher than leveling when opens the door	1. Board installation position is not accurate; 2. Photoelectric distance is inaccurate;	1. Adjust the board position of each floor when elevator runs slowly. 2. Adjust the up and down photoelectric spacing and the value of UP level and DOWN level in M3310
23	The elevator runs to the target floor, but it is lower than leveling when opens the door	1. Board installation position is not accurate; 2. Photoelectric distance is inaccurate;	1. Adjust the board position of each floor when elevator runs slowly 2. Adjust the up and down photoelectric spacing
24	The elevator shakes in high frequency when starts	1 Adjustment of pre-torque is inaccurate. 2, The position of main machine is inaccurate;	1. Check value of KP, KI in M335. 2. Check the drive parameters of M3314 and the drive size of M331-7412
25	When the elevator starts, noise of main machine is very large	1, The parameter setting of main machine and inverter, such as inertia, inductance, resistance and carrier, is incorrect. 2, KP and KI, the compensation parameter of torque is not appropriate; 3, the brake is not completely open;	1. Check the main machine parameter of M3314 and drive size of M331-7412. Confirm the Inertia of M3311 kg/m ² , Ld0, Lq0, Ld, Lq of M331-4615 and switch frequency of M331-3698. 2. Confirm KP and KI value of M335 3. Confirm the brake spacing is too small

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26	Noise of main machine is very large when elevator is running	1, The parameter setting of inverter is incorrect, such as the setting of inertia and inductance is too large; 2, setting of carrier frequency is unreasonable;	Check the main machine parameter of M3314 and drive size of M331-7412. Confirm the Inertia of M3311 kg/m2, Ld0, Lq0, Ld, Lq of M331-4615and switch frequency of M331-3698	
27	When the elevator stops, the noise is very large	1, The parameter setting of inverter is incorrect, such as the setting of inertia and inductance is too large; 2, setting of carrier frequency is unreasonable	Check the main machine parameter of M3314 and drive size of M331-7412. Confirm the Inertia of M3311 kg/m2, Ld0, Lq0, Ld, Lq of M331-4615and switch frequency of M331-3698	
28	电梯以半速运行 The elevator runs at half speed	1, Did not self learning 2, parameter setting is inaccurate; (rope ratio, diameter of tractive sheave is inaccurate, Velocity normal is inaccurate)	1. Check whether the height of each floor and travel switch of M314 is the same to reality 2. Check the rope ratio, tractive sheave diameter and Velocity normal of M3319 to be consistent with reality	
Door fault				
No.	Fault description	The possible reasons	methods of identification	
1	Elevator opens the door not on the target floor, but the next one. It can't run after tring three times	1, the door machine is abnormal, so can't open door completely; 2, There is DOL signal all the time	1. After the door machine receiving the DO signal, the door can't open or can't completely open as it is stuck 2. The DOL of M112 is capitalized in condition that the door doesn't open completely in fact.	
2	电梯运行到目标层不开门 Elevator doesn't open the door on the target floor	1. The DDO state; 2. Door machine type of the control panel is not set correctly; 3.The system faults after stopping, so door can't open 4. The wiring of door machine is wrong 5. Parameter setting of door	1. door status of M111 is DDO 2. DOOR value of M1315, set the relay control mode to 5, communication mode l to 12 3. check the drive fault code of M321 and logic fault of M121 4. Check the signal wiring of door opening 5. Check the parameter of door machine	

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		machine faults, doesn't accept system instruction;	
3	Elevator opens and closes the door all the time on the target floor	1, the signal of light curtain repeats action 2, The address setting of light curtain, light eyes and safety edges is incorrect, it causes that signal always be there.	1. The LRD of M112 repeat capitalizing and lowercasing 2. In M132, check the address of EDP, SGS and LRD
4	When arrives at the target floor, elevator keeps opening the door, and the buzzer outputs.	1, signal of light curtain, light eyes and safety board is there all the time; 2, overload signal always be there.	1. The DOS, LRD and EDP of M112 is capitalized all the time 2. The LWO of M112 always is capitalized
5	The door does not open fully, only opens a half.	1, door machine can't move as is jammed; 2, conversion board of door machine is abnormal; 3, parameter setting of LDR error; 4, in the process of the opening or closing door, elevator goes into maintenance mode;	1. Check whether the door machine is jammed 2. Check signal of open/close door in conversion board of door machine. If signal is open/close fully, check whether signal of light curtain acts normal. 3. LDR of M1315 = 1 4. In the process of the elevator opens/closes door, check whether the ERO of M112 is capital
6	The elevator does not close completely, but in a state of half closed	1, door machine loses power 2, door machine is jammed; 3, in the process of the opening or closing door, elevator goes into maintenance mode	1. Go to car roof and check whether door machine loses power 2. Check whether hall door is jammed on the problem floor 3. In the process of the elevator opens/closes door, check whether the ERO of M112 is capital
7	The hall door collides as door closes too hard	The parameters setting of door machine is wrong	Check the parameters of door machine

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8	The elevator can run when door opens	The door lock is shortened	DFC and DW of M112 is capital when door opens	
Inside and outside calls failure				
No.	Fault description	The possible reasons	methods of identification	
1	Buttons of control box is not bright, but have a response	1, the button light is damaged; 2, There is a problem with circuit wiring of button light; 3, the RS5 is damaged;	1. Change a better button to solve the problem 2. Changing a better button can't solve the problem 3. Change the RS5	
2	Buttons of control box is not bright, and no response	1, communication failures, communication between RS5 and RSL of mainboard faults; 2, incorrect address; 3, setting of floor response faults; 4, If it is only one button does not respond, maybe button is damaged or wiring problem in the COP	1. Check the communication connection of RSL 2. M132 check the address of floor button 3. M1331 check the floor setting of response function 4. Change a better button	
3	Button in COP is bright, but no response	1, The elevator faults in other floor, such as someone or something blocks light curtain for a long time; 2, Door machine faults, it can't close door fully or no lock signal, so the elevator can't run;	1. Check whether someone or something blocks the door on the other floors 1. In M112, DCL is lowercase, DFC or DW is lowercase	
4	The elevator runs to the wrong floor	1, check whether there is a blind floor; 2, address setting of I/O error;	1. Confirm whether there is a virtual floor setting of M1331 2. Check whether the IO address of M132 is the same as the one on schematic diagram	
5	The button flashes when elevator runs	1, Configuration of terminal absorbing panel faults; 2, RSL signal is interfered	1. Check whether some parts of the terminal absorbing panel are missing or burned 2. Confirm that circuit of RSL signal and strong electric is isolated	

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		by strong electricity 3, Setting of I/O address conflicts;	3. Compare the IO address of M132 with the one on schematic diagram 4. Check the jumpers of J8, J10, J15 and J16 on mainboard. If Parallel elevators share the hall call or two of group control elevators share the hall call, should cancel the above 4 jumpers.	
display fault				
1	Displayed floor is not correct	1, Setting of display is incorrect; 2, There is a blind floor, but hasn't set the display of the blind floor	Check the setting of floor display of M134	
2	The display has no image, but still can make call as normal	1, plugin (H) not connect or poor connection; 2, The display is bad;	1. Check the plugin (H) on the display 2. After changing a good display, it is normal	
3	Display with a splash screen	The display is bad;	After changing a good display, it is normal	
4	Display with missing codes	The display is bad	After changing a good display, it is normal	
5	Always display full load	1, the signal of full load inputs all the time; 2, The display is bad 3, The address setting of full load light is other I/O	1. Whether the address of LNS in M132 is input by force 2. Change the display 3. Whether LNSL address of M132 is the same as other IO	
6	Always display the fire fighting	1, the signal of fire fighting inputs all the time 2, The display is bad 3, The address setting of fire fighting light is other I/O	1, Whether the address of EFK in M132 is input by force 2. Change the display 3, Whether FSL address of M132 is the same as other IO	
7	Always display over load	1, the signal of over load inputs all the time; 2, The display is bad; 3, The address setting of over load light is other I/O	1, Whether the address of LWS in M132 is input by force 2. Change the display 3. Whether OLS address of M132 is the same as other IO	
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8	The showed floor information is earlier than the actual floor	1, Acceleration and deceleration settings of the drive is too small	The value setting of Accelera normal and Decelera normal in M3319 is too small	